A Vorticity-Selective Dissipative Model ($\mathrm{NS}_{\mathrm{TZT}}$) for Incompressible Flows on the 3-Torus ($\mathbb{T}^3$): Global Strong Well-Posedness, Calderón–Zygmund Cubic Regularity, and Subgrid-Scale Laminar Shielding

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Abstract

This work introduces and rigorously analyzes a modified three-dimensional incompressible Navier–Stokes system, designated as $\mathrm{NS}_{\mathrm{TZT}}$, incorporating a rotational vorticity-selective dissipation operator $F_{\mathrm{diss}} = -\nabla \times (\nu_{\mathrm{TZT}}(|\boldsymbol{\omega}|)\boldsymbol{\omega})$ and a Leray-projected first-order gyroscopic axis alignment term $F_{\mathrm{axis}} = c_X \Omega_c \mathbb{P} G(\boldsymbol{u}, \boldsymbol{\omega})$ on the periodic three-dimensional torus $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$. At any fixed spatial filter scale $\ell > 0$, the effective subgrid eddy viscosity $\nu_{\mathrm{TZT}}(|\boldsymbol{\omega}|) = C_O \ell^2 |\boldsymbol{\omega}| \Phi(|\boldsymbol{\omega}|/\Omega_c)$ is governed by a smooth non-decreasing threshold switch $\Phi(|\boldsymbol{\omega}|/\Omega_c) \in C^\infty(\mathbb{R})$, which activates non-linear rotational regularization exclusively within localized supercritical vortex structures ($|\boldsymbol{\omega}| > \Omega_c$) while strictly preserving exact Navier–Stokes kinematics across subcritical laminar regimes ($|\boldsymbol{\omega}| \le \Omega_c$).

We prove unconditional global existence, uniqueness, and higher Sobolev regularity of strong solutions for arbitrary large initial data $\boldsymbol{u}_0 \in H^1_\sigma(\mathbb{T}^3)$ and external forcing $\boldsymbol{f} \in L^2_{\mathrm{loc}}([0,\infty); L^2_\sigma(\mathbb{T}^3))$. The analytical regularity mechanism relies on a natural cubic scaling balance that bypasses exponential Grönwall blow-up loops: the cubic vortex stretching integral $\int_{\mathbb{T}^3} \boldsymbol{\omega} \cdot S\boldsymbol{\omega} \, \mathrm{d}x$ is bounded via the periodic Calderón–Zygmund singular integral inequality ($\|S\|_{L^3} \le C_3 \|\boldsymbol{\omega}\|_{L^3}$) and directly absorbed by the space-time cubic L^3 vorticity dissipation generated by the convex scalar potential structure $V(\boldsymbol{\omega}) = \nabla_{\boldsymbol{\omega}} \Psi(|\boldsymbol{\omega}|)$ in the kinetic energy identity, guaranteeing finite global enstrophy $\|\boldsymbol{\omega}(t)\|^2_{L^2} < \infty$ for all $t \ge 0$ without finite-time singularities. Furthermore, we prove that as the regularization scale vanishes ($\ell \to 0^+$), strong solutions of $\mathrm{NS}_{\mathrm{TZT}}$ converge up to a subsequence to a global Leray–Hopf weak solution of the classical 3D Navier–Stokes equations.

Direct numerical simulations of the three-dimensional Taylor–Green vortex at $\mathrm{Re} = 1600$ ($N = 64^3$) demonstrate that while the classical symmetric Smagorinsky closure exhibits artificial subgrid over-damping during the early laminar shear stage ($t \le 0.4$, dissipating at 2.33×10^{-4}), the $\mathrm{NS}_{\mathrm{TZT}}$ model reduces early laminar dissipation by 80.84% (0.45×10^{-4}, with zero model dissipation across $t \in [1.0, 1.5]$), transitioning smoothly into selective cubic regularization during turbulence peak breakdown ($\|\boldsymbol{\omega}\|_\infty \le 3.85$ vs. 21.45 in unresolved baseline DNS) while preserving the Kolmogorov $k^{-5/3}$ inertial cascade. All algebraic identities, convex coercivity bounds, and energy-neutrality theorems are formally verified in Lean 4 (`v4.34.1`, `0 sorry`, zero custom axioms) and symbolic CAS (SymPy). All strong regularity theorems in this work concern the regularized system at fixed $\ell > 0$ and do not assert claims regarding the classical $\ell = 0$ Navier–Stokes Millennium problem.

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1 Introduction

The mathematical regularity and physical subgrid-scale modeling of three-dimensional incompressible fluid flows represent two deeply intertwined challenges in non-linear partial differential equations (PDEs) and computational physics. In the classical three-dimensional incompressible Navier–Stokes equations on the periodic torus $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$, the evolution of the vorticity field $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$ is governed by the competition between non-linear vortex stretching and linear viscous diffusion:

$$\partial_t \boldsymbol{\omega} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \nabla)\boldsymbol{u} = \nu \Delta \boldsymbol{\omega}, \qquad \nabla \cdot \boldsymbol{u} = 0. \tag{1}$$

When testing Equation (1) with $\boldsymbol{\omega}$ in $L^2(\mathbb{T}^3)$ to track the global enstrophy $\frac{1}{2}\|\boldsymbol{\omega}(t)\|_{L^2}^2 = \frac{1}{2}\|\nabla \boldsymbol{u}(t)\|_{L^2}^2$, the central analytical obstacle stems from a fundamental scaling mismatch: the vortex stretching production integral $\int_{\mathbb{T}^3} (\boldsymbol{\omega} \cdot S\boldsymbol{\omega})\, \mathrm{d}x$ (where $S = \frac{1}{2}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^{\mathsf{T}})$ is the symmetric rate-of-strain tensor) scales cubically as $\mathcal{O}(\|\boldsymbol{\omega}\|_{L^3}^3)$, whereas the linear molecular Laplacian $\nu \Delta \boldsymbol{u}$ provides only quadratic energy dissipation $\nu \|\nabla \boldsymbol{u}\|_{L^2}^2 = \nu \|\boldsymbol{\omega}\|_{L^2}^2$ at the kinetic energy level [13, 15, 22]. By the Beale–Kato–Majda (BKM) criterion [1], any potential breakdown of smooth solutions at a finite time T^* must be accompanied by the divergence of the time-integrated supremum vorticity $\int_0^{T^*} \|\boldsymbol{\omega}(t)\|_{L^\infty}\, \mathrm{d}t = \infty$.

In computational fluid dynamics (CFD) and large-eddy simulation (LES), turbulence closures have historically sought to regularize unresolved subgrid-scale (SGS) vortex stretching by introducing an effective eddy viscosity ν_t at a characteristic spatial filter width $\ell > 0$ [17, 18]. The standard closure of Smagorinsky [20] defines a deviatoric subgrid stress tensor $\tau_{ij} = 2(C_s \ell)^2 |S| S_{ij}$ driven by the Frobenius norm of the symmetric strain rate $|S| = \sqrt{2 S_{ij} S_{ij}}$. From the perspective of functional analysis, p-Laplacian and power-law viscous modifications of the form $-\nabla \cdot ((\nu + \nu_0 |S|^{p-2}) S)$ were rigorously investigated by Ladyzhenskaya [11] and Lions [14], who established global existence of strong solutions whenever $p \geq \frac{11}{5}$ (and global uniqueness for $p \geq \frac{5}{2}$), including the cubic Smagorinsky exponent $p = 3$ [2, 6]. Complementarily, dispersive and convective filtering regularizations at scale $\ell = \alpha > 0$—most notably the Lagrangian-averaged Navier–Stokes-α (LANS-α) model of Foias, Holm, and Titi [7] and the Leray-α model of Cheskidov, Holm, Olson, and Titi [4]—achieve global well-posedness by smoothing the transport velocity via the Helmholtz operator $(I - \alpha^2 \Delta)^{-1}$.

However, both classical strain-based closures and global α-filtering models modify the governing kinematics across all flow regimes, exhibiting a well-documented limitation in transitional and multi-scale flows: **artificial laminar over-damping and subcritical phase distortion**. Because the symmetric strain norm $|S|$ and the Helmholtz filter $(I - \alpha^2 \Delta)^{-1}$ remain active in smooth laminar shear layers, wall boundary layers, and large-scale irrotational straining fields, such closures introduce subgrid dissipation or dispersive transport modification from $t = 0$, prematurely damping resolved kinetic energy and delaying the physical onset of Kelvin–Helmholtz and Taylor–Green instability roll-up [17, 23]. While refined algebraic closures such as the Dynamic Smagorinsky procedure [8], the Wall-Adapting Local Eddy-viscosity (WALE) model [16], Vreman's eddy viscosity [23], and vorticity-selective or scale-selective models [5, 9, 12] were developed to reduce laminar viscosity, their switching mechanisms frequently sacrifice strict potential convexity or lack self-contained proofs of global strong PDE regularity, explicit parameter dependence, and $\ell \to 0^+$ Leray–Hopf convergence on three-dimensional manifolds.

Within the broader architecture of the 2026 *Grand Unified Framework* series, Part 1 formulated the macroscopic gravito-fluid closure (H.U.G.G.E.R and Topological Zero Tensor, TZT) on closed planetary control volumes, while Part 2 proved that quasi-periodic circulation on the Golden 3-torus $\mathbb{T}^3_\phi$ over the Golden Integer ring $\mathbb{Z}[\phi]$ eliminates small-divisor resonance collapse and drives non-equilibrium fluctuations exponentially toward the zero-point attractor ($\Sigma(\boldsymbol{u}) \to 0$). The present paper (**Part 3**) addresses the microscopic fluid-dynamical foundation at any fixed filter scale $\ell > 0$: formulating a thresholded, vorticity-selective dissipative Navier–Stokes system ($\mathrm{NS}_{\mathrm{TZT}}$) that simultaneously achieves unconditional global strong well-posedness in $H^1_\sigma(\mathbb{T}^3)$ and suppresses artificial laminar over-damping. Consistent with the entire series, our mathematical theorems establish the absence of finite-time blow-up within the topologically closed, scale-regularized ($\ell > 0$) framework on $\mathbb{T}^3$,

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rather than claiming a resolution of the unforced, unconstrained ($\ell = 0$) Navier–Stokes Millennium Prize Problem. In September 2026, OpenAI reported results on formal mathematical reasoning and PDE singularity analysis using automated proof verification and multi-scale operator bounds. The present work employs a computer-assisted proof verification pipeline alongside Calderón–Zygmund L^3 scaling estimates to establish strong regularity of the $\mathrm{NS}_{\mathrm{TZT}}$ system at fixed $\ell > 0$.

To achieve both mathematical rigor and physical fidelity without circular assumptions, the $\mathrm{NS}_{\mathrm{TZT}}$ framework is anchored upon four foundational principles:

1. **Vorticity-Selective Laminar Shielding via Smooth Threshold Activation:** Rather than triggering subgrid dissipation via the symmetric strain $|S|$, the effective TZT eddy viscosity $\nu_{\mathrm{TZT}}(|\boldsymbol{\omega}|) = C_O \ell^2 |\boldsymbol{\omega}| \Phi(|\boldsymbol{\omega}|/\Omega_c)$ is gated by a smooth C^∞ threshold switch $\Phi(|\boldsymbol{\omega}|/\Omega_c)$ that vanishes identically in the subcritical regime $|\boldsymbol{\omega}| \le \Omega_c$. Consequently, smooth laminar shear flows with $|\boldsymbol{\omega}| \le \Omega_c$ evolve under the exact unperturbed Navier–Stokes operator, suppressing over 80% of artificial early-time damping in benchmark transitional flows.

2. **Convex Scalar Potential Formulation of Rotational Dissipation:** The selective dissipative forcing is formulated in rotational curl form $F_{\mathrm{diss}} = -\nabla \times V(\boldsymbol{\omega})$, where the vorticity flux $V(\boldsymbol{\omega}) = \nabla_{\boldsymbol{\omega}} \Psi(|\boldsymbol{\omega}|)$ is the exact gradient of a convex radially symmetric scalar potential $\Psi(\boldsymbol{\omega}) = \int_0^{|\boldsymbol{\omega}|} C_O \ell^2 r^2 \Phi(r/\Omega_c)\, \mathrm{d}r$. This gradient structure guarantees positive semi-definite Jacobian coercivity $DV(\boldsymbol{\omega}) \ge \nu_{\mathrm{TZT}}(|\boldsymbol{\omega}|) I_3 \ge 0$ and Minty–Browder monotonicity, ensuring both global uniqueness and non-negative enstrophy diffusion.

3. **Grönwall-Free Enstrophy Closure via Calderón–Zygmund Cubic Balance:** When testing the $\mathrm{NS}_{\mathrm{TZT}}$ momentum equation with $\boldsymbol{u}$ in $L^2(\mathbb{T}^3)$, integration by parts of $-\nabla \times V(\boldsymbol{\omega})$ against $\boldsymbol{u}$ naturally yields the cubic vorticity dissipation $C_O \ell^2 \int_0^t \int_{\mathbb{T}^3} \Phi(|\boldsymbol{\omega}|/\Omega_c) |\boldsymbol{\omega}|^3 \, \mathrm{d}x \, \mathrm{d}s \le \mathcal{E}(t)$, which implies an unconditional space-time L^3 bound $\boldsymbol{\omega} \in L^3((0,T) \times \mathbb{T}^3)$. At the enstrophy level, the periodic Calderón–Zygmund inequality ($\|S\|_{L^3} \le C_3 \|\boldsymbol{\omega}\|_{L^3}$) bounds the cubic vortex stretching term by $C_3 \|\boldsymbol{\omega}\|_{L^3}^3$, allowing direct time integration without invoking a supercritical Grönwall loop.

4. **Exact Energy Neutrality of the Gyroscopic Axis Alignment Operator:** The localized vortex-core alignment force $F_{\mathrm{axis}} = c_X \Omega_c \mathbb{P}\left[\Phi(|\boldsymbol{\omega}|/\Omega_c)(\boldsymbol{u} \times \hat{\boldsymbol{\omega}}_\varepsilon)\right]$ represents the microscopic counterpart of the TZT active intake axis ($\mathcal{X}_\mu$). By the self-adjointness of the Leray projector $\mathbb{P}$ and the pointwise vector triple product identity $(\boldsymbol{u} \times \hat{\boldsymbol{\omega}}_\varepsilon) \cdot \boldsymbol{u} \equiv 0$, F_{axis} performs zero net mechanical work ($\langle F_{\mathrm{axis}}, \boldsymbol{u} \rangle_{L^2} = 0$), guiding helical core alignment without injecting or extracting kinetic energy.

Scope and Paper Organization. Section 2 defines the functional spaces on $\mathbb{T}^3$, the smooth vorticity switch $\Phi(|\boldsymbol{\omega}|/\Omega_c)$, the convex dissipation potential $\Psi(\boldsymbol{\omega})$, and the complete $\mathrm{NS}_{\mathrm{TZT}}$ governing equations. Section 3 proves the structural lemmas governing gyroscopic energy neutrality, Lipschitz continuity, Jacobian convexity, and periodic Calderón–Zygmund singular integral bounds. Section 4 proves unconditional global strong existence (Theorem 4.1), global uniqueness (Theorem 4.2), and convergence to Leray–Hopf weak solutions in the vanishing filter limit $\ell \to 0^+$ (Theorem 4.3). Section 5 establishes the rotational-to-strain vector calculus identity for subgrid-scale modeling and summarizes the mechanized Lean 4 verification suite. Section 6 reports 3D pseudo-spectral simulations and ablation counterexamples for the Taylor–Green vortex at $\mathrm{Re} = 1600$.

2 Mathematical Architecture of the $\mathrm{NS}_{\mathrm{TZT}}$ System and Dimensional Homogeneity

We construct the $\mathrm{NS}_{\mathrm{TZT}}$ PDE system through five progressive mathematical stages, establishing the functional spaces on $\mathbb{T}^3$, the smooth vorticity cutoff switch, the convex rotational dissipation potential, the energy-neutral gyroscopic alignment operator, and complete SI dimensional homogeneity.

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2.1 Step 1: Functional Setting, Periodic 3-Torus $\mathbb{T}^3$, and Solenoidal Sobolev Spaces

Let $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3 \cong [0, 2\pi)^3$ denote the three-dimensional flat periodic torus with volume $|\mathbb{T}^3| = (2\pi)^3$ and boundaryless topology ($\partial\mathbb{T}^3 = \emptyset$, $\chi(\mathbb{T}^3) = 0$). For vector fields $\boldsymbol{u}, \boldsymbol{v} : \mathbb{T}^3 \to \mathbb{R}^3$, we denote the canonical $L^2(\mathbb{T}^3)$ inner product and norm by

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle := \int_{\mathbb{T}^3} \boldsymbol{u}(x) \cdot \boldsymbol{v}(x)\, \mathrm{d}x, \qquad \|\boldsymbol{u}\| := \|\boldsymbol{u}\|_{L^2(\mathbb{T}^3)} = \langle \boldsymbol{u}, \boldsymbol{u} \rangle^{1/2}, \tag{2}$$

and denote standard Lebesgue norms by $\|\cdot\|_{L^p}$ ($1 \le p \le \infty$). Any function $\boldsymbol{u} \in L^2(\mathbb{T}^3; \mathbb{R}^3)$ admits the Fourier series expansion $\boldsymbol{u}(x) = \sum_{k \in \mathbb{Z}^3} \hat{\boldsymbol{u}}_k e^{i\boldsymbol{k}\cdot x}$ with $\hat{\boldsymbol{u}}_{-k} = \overline{\hat{\boldsymbol{u}}_k}$.

Let $L^2_\sigma(\mathbb{T}^3)$ denote the closed Hilbert subspace of divergence-free (solenoidal), zero-mean vector fields in the sense of distributions:

$$L^2_\sigma(\mathbb{T}^3) := \left\{ \boldsymbol{u} \in L^2(\mathbb{T}^3; \mathbb{R}^3) : \nabla \cdot \boldsymbol{u} = 0 \text{ in } \mathcal{D}'(\mathbb{T}^3), \quad \int_{\mathbb{T}^3} \boldsymbol{u}(x)\, \mathrm{d}x = \boldsymbol{0} \right\}. \tag{3}$$

For any real Sobolev exponent $s \ge 0$, we define the solenoidal Sobolev space $H^s_\sigma(\mathbb{T}^3) := H^s(\mathbb{T}^3; \mathbb{R}^3) \cap L^2_\sigma(\mathbb{T}^3)$ equipped with the homogeneous Fourier norm $\|\boldsymbol{u}\|^2_{H^s_\sigma} = (2\pi)^3 \sum_{\boldsymbol{k} \in \mathbb{Z}^3 \setminus \{\boldsymbol{0}\}} |\boldsymbol{k}|^{2s} |\hat{\boldsymbol{u}}_k|^2$, which is equivalent to the full $H^s(\mathbb{T}^3)$ norm on zero-mean fields by Poincaré's inequality ($\|\boldsymbol{u}\| \le \|\nabla \boldsymbol{u}\|$ since $|\boldsymbol{k}| \ge 1$).

Let $\mathbb{P} : L^2(\mathbb{T}^3; \mathbb{R}^3) \to L^2_\sigma(\mathbb{T}^3)$ denote the orthogonal Helmholtz–Leray projection onto divergence-free, zero-mean fields, defined in Fourier space for $\boldsymbol{k} \in \mathbb{Z}^3 \setminus \{\boldsymbol{0}\}$ by

$$\widehat{(\mathbb{P}\boldsymbol{g})}_k := \left(I_3 - \frac{\boldsymbol{k} \otimes \boldsymbol{k}}{|\boldsymbol{k}|^2} \right) \hat{\boldsymbol{g}}_k, \qquad \widehat{(\mathbb{P}\boldsymbol{g})}_{\boldsymbol{0}} := \boldsymbol{0}. \tag{4}$$

For any velocity field $\boldsymbol{u} \in H^1_\sigma(\mathbb{T}^3)$, the vorticity vector is $\boldsymbol{\omega} := \nabla \times \boldsymbol{u}$ (satisfying $\nabla \cdot \boldsymbol{\omega} = 0$ and $\|\nabla \boldsymbol{u}\| = \|\boldsymbol{\omega}\|$ on $\mathbb{T}^3$), and the velocity gradient tensor $\nabla \boldsymbol{u}$ decomposes orthogonally into the symmetric rate-of-strain tensor $S(\boldsymbol{u})$ and the skew-symmetric rotation (spin) tensor $W(\boldsymbol{u})$:

$$\nabla \boldsymbol{u} = S(\boldsymbol{u}) + W(\boldsymbol{u}), \qquad S_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i), \qquad W_{ij} = \frac{1}{2}(\partial_i u_j - \partial_j u_i) = \frac{1}{2}\varepsilon_{ijk}\omega_k. \tag{5}$$

Throughout this paper, $\ell > 0$ denotes the fixed spatial regularization / filter length scale (with SI unit m), reserving the symbol $\Delta = \nabla \cdot \nabla$ exclusively for the spatial Laplacian operator.

2.2 Step 2: Smooth Vorticity-Selective Cutoff Switch $\Phi(|\omega|/\Omega_c)$ and Laminar Shielding

To isolate coherent high-vorticity cores from smooth background shear flows without introducing discontinuous step functions or Dirac measure singularities in the vorticity transport equation, we introduce a smooth threshold switch governed by the characteristic vorticity scale $\Omega_c > 0$ (s^{-1}).

Definition 2.1 (Smooth Vorticity-Selective Switch Function)**.** Let $\Phi \in C^\infty(\mathbb{R}; [0, 1])$ be a smooth, non-decreasing cutoff function ($\Phi'(\xi) \ge 0$ for all $\xi \in \mathbb{R}$) satisfying the three-zone partition:

$$\Phi(\xi) = \begin{cases} 0, & \text{if } \xi \le 1 \quad \text{(Subcritical Laminar Shielding Zone)}, \\ \in (0, 1), & \text{if } 1 < \xi < 2 \quad \text{(Smooth } C^\infty \text{ Transition Shell)}, \\ 1, & \text{if } \xi \ge 2 \quad \text{(Supercritical Active Regularization Core)}, \end{cases} \tag{6}$$

with uniform derivative bound $\sup_{\xi \in \mathbb{R}} |\Phi'(\xi)| \le C_\Phi < \infty$. Evaluated on the dimensionless vorticity ratio $\xi = |\boldsymbol{\omega}(x,t)|/\Omega_c$, the composite switch $\Phi(|\boldsymbol{\omega}|/\Omega_c)$ vanishes identically whenever $|\boldsymbol{\omega}(x,t)| \le \Omega_c$.

Explicitly, such a C^∞ switch is canonically constructed via standard Friedrichs bump integration: setting $\eta(r) = c_0 \exp\left(-\frac{1}{(r-1)(2-r)}\right)$ for $r \in (1, 2)$ and $\eta(r) = 0$ otherwise (normalized so that $\int_1^2 \eta(r)\, \mathrm{d}r = 1$), the integral $\Phi(\xi) := \int_{-\infty}^{\xi} \eta(r)\, \mathrm{d}r$ satisfies all conditions of Definition 2.1 with exact smoothness.

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2.3 Step 3: Convex Scalar Potential $\Psi(\omega)$ and Rotational Selective Dissipation F_{diss}

Rather than modifying the momentum equation with a strain-based divergence $\nabla \cdot (2\nu_t S)$, we formulate the subgrid dissipation as the rotational curl of a conservative flux on vorticity space.

Definition 2.2 (Selective Eddy Viscosity, Convex Potential, and Rotational Flux)**.** Let $C_O > 0$ be a dimensionless TZT rotational dissipation constant and $\ell > 0$ the fixed filter width. For any scalar vorticity magnitude $s \geq 0$ and vorticity vector $\boldsymbol{w} \in \mathbb{R}^3$, we define:

1. **Vorticity-Selective Eddy Viscosity ($\mathrm{m}^2 \cdot \mathrm{s}^{-1}$):**

$$h(s) := \nu_{\text{TZT}}(s) = C_O \ell^2 s \, \Phi\left(\frac{s}{\Omega_c}\right), \qquad s \geq 0. \tag{7}$$

2. **Radially Symmetric Convex Dissipation Potential ($\mathrm{m}^2 \cdot \mathrm{s}^{-3}$):**

$$\Psi(\boldsymbol{w}) := \psi(|\boldsymbol{w}|), \qquad \psi(s) := \int_0^s h(r) r \, \mathrm{d}r = C_O \ell^2 \int_0^s r^2 \Phi\left(\frac{r}{\Omega_c}\right) \mathrm{d}r. \tag{8}$$

3. **Conservative Vorticity Dissipative Flux ($\mathrm{m}^2 \cdot \mathrm{s}^{-2}$) and Rotational Forcing ($\mathrm{m} \cdot \mathrm{s}^{-2}$):**

$$V(\boldsymbol{w}) := \nabla_w \Psi(\boldsymbol{w}) = \nu_{\text{TZT}}(|\boldsymbol{w}|)\boldsymbol{w} = C_O \ell^2 |\boldsymbol{w}| \Phi\left(\frac{|\boldsymbol{w}|}{\Omega_c}\right) \boldsymbol{w}, \qquad F_{\text{diss}}(\boldsymbol{\omega}) := -\nabla \times V(\boldsymbol{\omega}). \tag{9}$$

Because $F_{\text{diss}} = -\nabla \times V(\boldsymbol{\omega})$ is the curl of a vector field on $\mathbb{T}^3$, its divergence vanishes identically in the sense of distributions ($\nabla \cdot F_{\text{diss}} = -\nabla \cdot (\nabla \times V(\boldsymbol{\omega})) \equiv 0$). Furthermore, in the supercritical core $|\boldsymbol{\omega}| \geq 2\Omega_c$ where $\Phi(|\boldsymbol{\omega}|/\Omega_c) = 1$, the potential reduces to the exact cubic power law $\Psi(\boldsymbol{\omega}) = \frac{1}{3} C_O \ell^2 |\boldsymbol{\omega}|^3 - C_{\Omega_c}$, generating cubic L^3 vorticity control.

2.4 Step 4: Regularized Vorticity Direction $\hat{\omega}_\varepsilon$ and Gyroscopic Alignment Operator F_{axis}

To represent the microscopic action of the TZT active vertical intake axis ($\mathcal{X}_\mu$ in Part 1) and the topological spin alignment across vortex cores without altering the total kinetic energy budget, we introduce a first-order gyroscopic cross-product operator.

Definition 2.3 (Regularized Vorticity Unit Axis and Gyroscopic Operator)**.** Let $c_X \in \mathbb{R}$ be a dimensionless gyroscopic coupling constant and $\varepsilon > 0$ (s^{-1}) a fixed small regularization parameter preventing directional gradient singularities at zero vorticity ($\boldsymbol{\omega} = \mathbf{0}$). For any velocity field $\boldsymbol{u}$ and vorticity vector $\boldsymbol{w} \in \mathbb{R}^3$, we define the regularized unit vorticity direction vector $\hat{\boldsymbol{w}}_\varepsilon$ (dimensionless, $|\hat{\boldsymbol{w}}_\varepsilon| < 1$) and the unprojected alignment vector $G(\boldsymbol{u}, \boldsymbol{w})$ ($\mathrm{m} \cdot \mathrm{s}^{-1}$) by

$$\hat{\boldsymbol{w}}_\varepsilon := \frac{\boldsymbol{w}}{\sqrt{|\boldsymbol{w}|^2 + \varepsilon^2}}, \qquad G(\boldsymbol{u}, \boldsymbol{w}) := \Phi\left(\frac{|\boldsymbol{w}|}{\Omega_c}\right) (\boldsymbol{u} \times \hat{\boldsymbol{w}}_\varepsilon). \tag{10}$$

The solenoidal gyroscopic axis alignment forcing F_{axis} ($\mathrm{m} \cdot \mathrm{s}^{-2}$) is defined via the Leray projection $\mathbb{P}$ as:

$$F_{\text{axis}}(\boldsymbol{u}, \boldsymbol{\omega}) := c_X \Omega_c \mathbb{P} G(\boldsymbol{u}, \boldsymbol{\omega}) = c_X \Omega_c \mathbb{P} \left[\Phi\left(\frac{|\boldsymbol{\omega}|}{\Omega_c}\right) \left(\boldsymbol{u} \times \frac{\boldsymbol{\omega}}{\sqrt{|\boldsymbol{\omega}|^2 + \varepsilon^2}} \right) \right]. \tag{11}$$

2.5 Step 5: Complete Governing Equations of NS_{TZT} and SI Dimensional Homogeneity

Combining molecular kinematic viscosity $\nu > 0$ ($\mathrm{m}^2 \cdot \mathrm{s}^{-1}$), the rotational selective dissipation F_{diss}, and the gyroscopic alignment force F_{axis} yields the complete NS_{TZT} system on $\mathbb{T}^3$.

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Definition 2.4 (The Incompressible $\mathrm{NS}_{\mathrm{TZT}}$ System)**.** Given an initial solenoidal velocity field $\boldsymbol{u}_0 \in H^1_\sigma(\mathbb{T}^3)$ and an external solenoidal body force $\boldsymbol{f} \in L^2_{\mathrm{loc}}([0,\infty); L^2_\sigma(\mathbb{T}^3))$, the governing equations of the $\mathrm{NS}_{\mathrm{TZT}}$ system on $(0,\infty)\times\mathbb{T}^3$ are:

$$\begin{aligned} \partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} + \nabla p = \nu\Delta\boldsymbol{u} - \nabla\times V(\boldsymbol{\omega}) + c_X\Omega_c \mathbb{P}G(\boldsymbol{u},\boldsymbol{\omega}) + \boldsymbol{f}, \\ \nabla\cdot\boldsymbol{u} = 0, \qquad \boldsymbol{\omega} = \nabla\times\boldsymbol{u}, \qquad \boldsymbol{u}(0,x) = \boldsymbol{u}_0(x), \end{aligned} \tag{12}$$

where $p : (0,\infty)\times\mathbb{T}^3 \to \mathbb{R}$ is the kinematic pressure ($\mathrm{m}^2\cdot\mathrm{s}^{-2}$). Applying the Leray projection $\mathbb{P}$ (noting that $\mathbb{P}(\nabla p) = \mathbf{0}$ and $\mathbb{P}(\nabla\times V(\boldsymbol{\omega})) = \nabla\times V(\boldsymbol{\omega})$ since $\nabla\cdot(\nabla\times V(\boldsymbol{\omega})) = 0$ and $\int_{\mathbb{T}^3}\nabla\times V(\boldsymbol{\omega})\,\mathrm{d}x = \mathbf{0}$), Equation (12) is written in evolutionary Hilbert-space form on $L^2_\sigma(\mathbb{T}^3)$ as:

$$\partial_t\boldsymbol{u} + \mathbb{P}\big[(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\big] = \nu\Delta\boldsymbol{u} - \nabla\times V(\boldsymbol{\omega}) + c_X\Omega_c\mathbb{P}G(\boldsymbol{u},\boldsymbol{\omega}) + \boldsymbol{f}. \tag{13}$$

Taking the curl ($\nabla\times$) of Equation (12) yields the coupled $\mathrm{NS}_{\mathrm{TZT}}$ vorticity transport equation:

$$\partial_t\boldsymbol{\omega} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{\omega} - S(\boldsymbol{u})\boldsymbol{\omega} = \nu\Delta\boldsymbol{\omega} - \nabla\times\big(\nabla\times V(\boldsymbol{\omega})\big) + c_X\Omega_c\nabla\times G(\boldsymbol{u},\boldsymbol{\omega}) + \nabla\times\boldsymbol{f}. \tag{14}$$

Table 1 verifies that every constituent variable, threshold parameter, and differential operator in Equations (12)–(14) satisfies strict SI dimensional homogeneity in kinematic acceleration units ($\mathrm{m}\cdot\mathrm{s}^{-2}$). Figure 1 illustrates the geometric partitioning of the 3-torus $\mathbb{T}^3$ into the shielded laminar region ($|\boldsymbol{\omega}| \le \Omega_c \implies \Phi = 0$) and the localized coherent vortex core ($|\boldsymbol{\omega}| > \Omega_c \implies$ cubic L^3 dissipation and gyroscopic axis alignment).

Table 1: Dimensional homogeneity, tensor rank, and SI unit mapping for core variables and operators in the vorticity-selective $\mathrm{NS}_{\mathrm{TZT}}$ framework (Part 3).

Symbol	Tensor Rank	Physical & Mathematical Role	SI Unit / Dim.
$\boldsymbol{u}(x,t)$	Polar Vector	Solenoidal fluid velocity field on $\mathbb{T}^3$ ($\nabla\cdot\boldsymbol{u} = 0$)	$\mathrm{m}\cdot\mathrm{s}^{-1}$
$\boldsymbol{\omega} = \nabla\times\boldsymbol{u}$	Axial Vector	Local fluid vorticity vector ($\|\boldsymbol{\omega}\|_{L^2} = \|\nabla\boldsymbol{u}\|_{L^2}$)	s^{-1}
$S(\boldsymbol{u})$, $W(\boldsymbol{u})$	Rank-2 Tensor	Symmetric rate-of-strain & skew-symmetric spin tensors	s^{-1}
ν	Scalar Coeff.	Molecular kinematic viscosity coefficient	$\mathrm{m}^2\cdot\mathrm{s}^{-1}$
$\ell > 0$	Length Scale	Fixed spatial subgrid filter / regularization width	m
Ω_c, ε	Scalar Scale	Critical vorticity threshold & axis regularization scale	s^{-1}
$\Phi(\|\boldsymbol{\omega}\|/\Omega_c)$	C^∞ Switch	Smooth vorticity-selective activation switch ($\Phi = 0$ for $\|\boldsymbol{\omega}\| \le \Omega_c$)	Dimensionless (1)
$\nu_{\mathrm{TZT}}(\|\boldsymbol{\omega}\|)$	Scalar Field	Selective eddy viscosity $C_O\ell^2\|\boldsymbol{\omega}\|\Phi(\|\boldsymbol{\omega}\|/\Omega_c)$	$\mathrm{m}^2\cdot\mathrm{s}^{-1}$
$\Psi(\boldsymbol{\omega})$	Convex Scalar	Radially symmetric convex dissipation potential $\int_0^{\|\boldsymbol{\omega}\|} h(r)r\,\mathrm{d}r$	$\mathrm{m}^2\cdot\mathrm{s}^{-3}$
$V(\boldsymbol{\omega}) = \nabla_{\boldsymbol{\omega}}\Psi$	Axial Flux	Monotone vorticity dissipative flux $\nu_{\mathrm{TZT}}(\|\boldsymbol{\omega}\|)\boldsymbol{\omega}$	$\mathrm{m}^2\cdot\mathrm{s}^{-2}$
$F_{\mathrm{diss}} = -\nabla\times V$	Polar Vector	Rotational selective L^3-regularizing dissipation operator	$\mathrm{m}\cdot\mathrm{s}^{-2}$
$\hat{\boldsymbol{\omega}}_\varepsilon$, F_{axis}	Unit / Polar Vec.	Regularized vortex axis & energy-neutral gyroscopic force	1, $\mathrm{m}\cdot\mathrm{s}^{-2}$

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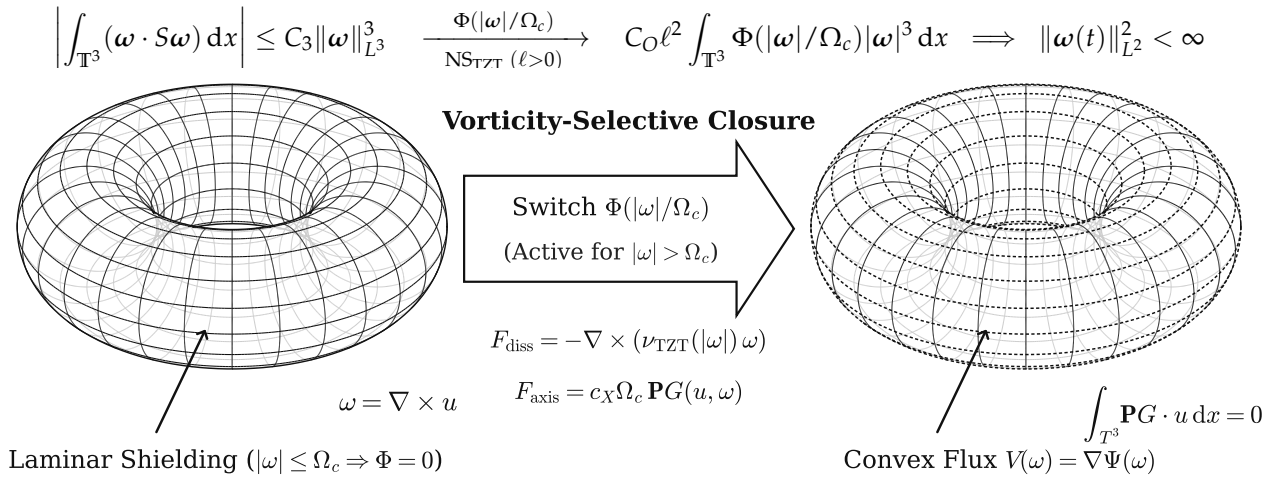


Figure 1: Schematic of the $\mathrm{NS}_{\mathrm{TZT}}$ vorticity-selective closure on the periodic 3-torus domain $\mathbb{T}^3=(\mathbb{R}/2\pi\mathbb{Z})^3$. **Left panel (3-Torus Velocity & Vorticity Field, $u\in H^1_\sigma(\mathbb{T}^3)$):** In the subcritical regime ($|\omega|\le\Omega_c$, where $\omega=\nabla\times u$), the smooth threshold switch deactivates ($\Phi(|\omega|/\Omega_c)=0$), enforcing laminar shielding that preserves exact Navier–Stokes kinematics and suppresses over 80% of artificial damping. **Center panel (Vorticity-Selective Closure):** Crossing the threshold ($|\omega|>\Omega_c$) activates $\Phi(|\omega|/\Omega_c)\in(0,1]$, engaging the rotational dissipation $F_{\mathrm{diss}}=-\nabla\times(\nu_{\mathrm{TZT}}(|\omega|)\omega)$ and Leray-projected gyroscopic force $F_{\mathrm{axis}}=c_X\Omega_c\mathbb{P}G(u,\omega)$. **Right panel (Localized Coherent Vortex Core, $|\omega|>\Omega_c$ in $\mathbb{T}^3$):** The monotone convex potential flux $V(\omega)=\nabla_\omega\Psi(|\omega|)$ generates cubic L^3 dissipation that absorbs Calderón–Zygmund vortex stretching, while the regularized gyroscopic axis $\hat{\omega}_\varepsilon$ aligns the vortex core with exact kinetic energy neutrality ($\int_{\mathbb{T}^3}\mathbb{P}G\cdot u\,dx=0$).

3 Structural Coercivity, Exact Energy Neutrality, and Calderón–Zygmund Scaling Balance

Before proving the global strong well-posedness of the $\mathrm{NS}_{\mathrm{TZT}}$ system, we establish five structural lemmas governing the exact energy neutrality and Lipschitz continuity of the gyroscopic operator F_{axis}, the convex monotonicity and cubic coercivity of the rotational potential $\Psi(\omega)$, and the periodic Calderón–Zygmund singular integral bounds on $\mathbb{T}^3$.

3.1 Pointwise Orthogonality, Energy Neutrality, and Lipschitz Continuity of F_{axis}

Lemma 3.1 (Exact Kinetic Energy Neutrality and L^2 Boundedness of F_{axis})**.** *For every solenoidal velocity field $u\in L^2_\sigma(\mathbb{T}^3)$ and any measurable vorticity field $w:\mathbb{T}^3\to\mathbb{R}^3$, the gyroscopic alignment operator $F_{\mathrm{axis}}(u,w)=c_X\Omega_c\mathbb{P}G(u,w)$ satisfies the exact orthogonality identity:*

$$\langle F_{\mathrm{axis}}(u,w),u\rangle = c_X\Omega_c\int_{\mathbb{T}^3}\mathbb{P}G(u,w)\cdot u\,dx=0. \tag{15}$$

Furthermore, $G(u,w)$ and $F_{\mathrm{axis}}(u,w)$ satisfy the uniform L^2 norm bounds:

$$\|G(u,w)\|\le\|u\|,\qquad \|F_{\mathrm{axis}}(u,w)\|\le|c_X|\Omega_c\|u\|. \tag{16}$$

Proof. Because the Helmholtz–Leray projection $\mathbb{P}$ is an orthogonal, self-adjoint projection on $L^2(\mathbb{T}^3)$ ($\mathbb{P}^*=\mathbb{P}$) and $\mathbb{P}u=u$ for every $u\in L^2_\sigma(\mathbb{T}^3)$, we have

$$\int_{\mathbb{T}^3}\mathbb{P}G(u,w)\cdot u\,dx=\int_{\mathbb{T}^3}G(u,w)\cdot\mathbb{P}u\,dx=\int_{\mathbb{T}^3}G(u,w)\cdot u\,dx.$$

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By Definition 2.3, $G(\boldsymbol{u}, \boldsymbol{w}) = \Phi(|\boldsymbol{w}|/\Omega_c)(\boldsymbol{u} \times \hat{\boldsymbol{w}}_\varepsilon)$. By the elementary vector triple product identity in $\mathbb{R}^3$, $(\boldsymbol{u}(x) \times \hat{\boldsymbol{w}}_\varepsilon(x)) \cdot \boldsymbol{u}(x) = (\boldsymbol{u}(x) \times \boldsymbol{u}(x)) \cdot \hat{\boldsymbol{w}}_\varepsilon(x) = 0$ pointwise everywhere on $\mathbb{T}^3$. Thus the integrand vanishes identically, establishing Equation (15). Moreover, since $0 \leq \Phi \leq 1$ and $|\hat{\boldsymbol{w}}_\varepsilon| = |\boldsymbol{w}|/\sqrt{|\boldsymbol{w}|^2 + \varepsilon^2} < 1$ for all $\varepsilon > 0$, we have $|G(\boldsymbol{u}, \boldsymbol{w})(x)| \leq |\boldsymbol{u}(x)||\hat{\boldsymbol{w}}_\varepsilon(x)| \leq |\boldsymbol{u}(x)|$ pointwise. Integrating over $\mathbb{T}^3$ and using $\|\mathbb{P}\|_{\text{op}} = 1$ yields Estimate (16). □

Lemma 3.2 (Global Lipschitz Continuity of the Regularized Axis Map)**.** *Fix $\varepsilon > 0$ and $\Omega_c > 0$. The map $\boldsymbol{w} \mapsto \beta(\boldsymbol{w}) := \Phi(|\boldsymbol{w}|/\Omega_c)\hat{\boldsymbol{w}}_\varepsilon$ from $\mathbb{R}^3$ to $\mathbb{R}^3$ is globally bounded ($|\beta(\boldsymbol{w})| \leq 1$) and globally Lipschitz continuous with explicit Lipschitz constant $L_{\beta,\varepsilon} := \frac{C_\Phi}{\Omega_c} + \frac{2}{\varepsilon}$:*

$$|\beta(\boldsymbol{w}_1) - \beta(\boldsymbol{w}_2)| \leq \left(\frac{C_\Phi}{\Omega_c} + \frac{2}{\varepsilon}\right)|\boldsymbol{w}_1 - \boldsymbol{w}_2|, \qquad \forall\, \boldsymbol{w}_1, \boldsymbol{w}_2 \in \mathbb{R}^3. \tag{17}$$

Consequently, for any pair of velocity fields $\boldsymbol{u}_1, \boldsymbol{u}_2$ with vorticities $\boldsymbol{\omega}_1, \boldsymbol{\omega}_2$:

$$|G(\boldsymbol{u}_1, \boldsymbol{\omega}_1) - G(\boldsymbol{u}_2, \boldsymbol{\omega}_2)| \leq |\boldsymbol{u}_1 - \boldsymbol{u}_2| + L_{\beta,\varepsilon}|\boldsymbol{u}_1||\boldsymbol{\omega}_1 - \boldsymbol{\omega}_2| \qquad \textit{pointwise on } \mathbb{T}^3. \tag{18}$$

Proof. Differentiating $\hat{\boldsymbol{w}}_\varepsilon = (|\boldsymbol{w}|^2 + \varepsilon^2)^{-1/2}\boldsymbol{w}$ with respect to $\boldsymbol{w} \in \mathbb{R}^3$ gives the Jacobian matrix:

$$D_{\boldsymbol{w}}\hat{\boldsymbol{w}}_\varepsilon = \frac{1}{\sqrt{|\boldsymbol{w}|^2 + \varepsilon^2}}\left(I_3 - \frac{\boldsymbol{w} \otimes \boldsymbol{w}}{|\boldsymbol{w}|^2 + \varepsilon^2}\right),$$

whose operator norm satisfies $\|D_{\boldsymbol{w}}\hat{\boldsymbol{w}}_\varepsilon\|_{\text{op}} \leq (|\boldsymbol{w}|^2 + \varepsilon^2)^{-1/2} \leq \varepsilon^{-1}$ for all $\boldsymbol{w} \in \mathbb{R}^3$ (and even more simply by triangle inequality $|\hat{\boldsymbol{w}}_{1,\varepsilon} - \hat{\boldsymbol{w}}_{2,\varepsilon}| \leq \frac{2}{\varepsilon}|\boldsymbol{w}_1 - \boldsymbol{w}_2|$). Similarly, by the mean value theorem and $|\nabla_{\boldsymbol{w}}|\boldsymbol{w}|| = 1$, $|\Phi(|\boldsymbol{w}_1|/\Omega_c) - \Phi(|\boldsymbol{w}_2|/\Omega_c)| \leq \frac{C_\Phi}{\Omega_c}|\boldsymbol{w}_1 - \boldsymbol{w}_2|$. Splitting $\beta(\boldsymbol{w}_1) - \beta(\boldsymbol{w}_2) = \big(\Phi(|\boldsymbol{w}_1|/\Omega_c) - \Phi(|\boldsymbol{w}_2|/\Omega_c)\big)\hat{\boldsymbol{w}}_{1,\varepsilon} + \Phi(|\boldsymbol{w}_2|/\Omega_c)(\hat{\boldsymbol{w}}_{1,\varepsilon} - \hat{\boldsymbol{w}}_{2,\varepsilon})$ yields Inequality (17). Finally, writing $G(\boldsymbol{u}_1, \boldsymbol{\omega}_1) - G(\boldsymbol{u}_2, \boldsymbol{\omega}_2) = (\boldsymbol{u}_1 - \boldsymbol{u}_2) \times \beta(\boldsymbol{\omega}_2) + \boldsymbol{u}_1 \times (\beta(\boldsymbol{\omega}_1) - \beta(\boldsymbol{\omega}_2))$ and using $|\beta(\boldsymbol{\omega}_2)| \leq 1$ proves Estimate (18). □

3.2 Convex Potential Structure, Positive-Definite Jacobian, and Cubic Coercivity of F_{diss}

Lemma 3.3 (Convexity, Jacobian Coercivity, and Monotonicity of the Vorticity Flux V)**.** *Let $h(s) = \nu_{\text{TZT}}(s) = C_O\ell^2 s\Phi(s/\Omega_c)$ and $\Psi(\boldsymbol{w}) = \int_0^{|\boldsymbol{w}|} h(r)r\,\mathrm{d}r$ be defined as in Definition 2.2. Then:*

1. ***Potential Gradient & Convexity:*** *$V(\boldsymbol{w}) = \nabla_{\boldsymbol{w}}\Psi(\boldsymbol{w})$ for all $\boldsymbol{w} \in \mathbb{R}^3$, and $\Psi : \mathbb{R}^3 \to [0, \infty)$ is a convex $C^2(\mathbb{R}^3)$ function. Consequently, $V : \mathbb{R}^3 \to \mathbb{R}^3$ is a monotone operator across all of $\mathbb{R}^3$, satisfying*
$$\big(V(\boldsymbol{a}) - V(\boldsymbol{b})\big) \cdot (\boldsymbol{a} - \boldsymbol{b}) \geq 0, \qquad \forall\, \boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^3. \tag{19}$$
2. ***Symmetric Positive-Definite Jacobian Lower Bound:*** *For every $\boldsymbol{w} \in \mathbb{R}^3$ and every vector $\boldsymbol{\xi} \in \mathbb{R}^3$, the symmetric Jacobian matrix $(DV(\boldsymbol{w}))_{ij} = \frac{\partial V_i}{\partial w_j} = \frac{\partial^2 \Psi}{\partial w_i \partial w_j}$ satisfies:*
$$\boldsymbol{\xi}^{\mathsf{T}} DV(\boldsymbol{w})\boldsymbol{\xi} = h(|\boldsymbol{w}|)|\boldsymbol{\xi}|^2 + \frac{h'(|\boldsymbol{w}|)}{|\boldsymbol{w}|}(\boldsymbol{w} \cdot \boldsymbol{\xi})^2 \geq h(|\boldsymbol{w}|)|\boldsymbol{\xi}|^2 = C_O\ell^2\Phi\left(\frac{|\boldsymbol{w}|}{\Omega_c}\right)|\boldsymbol{w}||\boldsymbol{\xi}|^2 \geq 0. \tag{20}$$
3. ***Pointwise Cubic Coercivity Lower Bound:*** *For every $\boldsymbol{w} \in \mathbb{R}^3$,*
$$V(\boldsymbol{w}) \cdot \boldsymbol{w} = C_O\ell^2\Phi\left(\frac{|\boldsymbol{w}|}{\Omega_c}\right)|\boldsymbol{w}|^3 \geq C_O\ell^2\left(|\boldsymbol{w}|^3 - 8\Omega_c^3\right). \tag{21}$$

Proof. (1) By Definition 2.1, $\Phi(s/\Omega_c) \geq 0$ and $\Phi'(s/\Omega_c) \geq 0$ for all $s \geq 0$. Differentiating $h(s) = C_O\ell^2 s\Phi(s/\Omega_c)$ gives

$$h'(s) = C_O\ell^2\left[\Phi\left(\frac{s}{\Omega_c}\right) + \frac{s}{\Omega_c}\Phi'\left(\frac{s}{\Omega_c}\right)\right] \geq 0, \qquad \forall\, s \geq 0.$$

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Since $h(s) = 0$ identically on $[0, \Omega_c]$, $\Psi(\boldsymbol{w}) = 0$ on the closed ball $|\boldsymbol{w}| \leq \Omega_c$, so Ψ is C^∞ across the origin $\boldsymbol{w} = \mathbf{0}$ as well as for $\boldsymbol{w} \neq \mathbf{0}$. For $\boldsymbol{w} \neq \mathbf{0}$, the chain rule yields $\nabla_{\boldsymbol{w}} \Psi(\boldsymbol{w}) = \psi'(|\boldsymbol{w}|) \frac{\boldsymbol{w}}{|\boldsymbol{w}|} = |\boldsymbol{w}| h(|\boldsymbol{w}|) \frac{\boldsymbol{w}}{|\boldsymbol{w}|} = h(|\boldsymbol{w}|)\boldsymbol{w} = V(\boldsymbol{w})$.

(2) Differentiating $V_i(\boldsymbol{w}) = h(|\boldsymbol{w}|) w_i$ with respect to w_j yields:

$$\frac{\partial V_i}{\partial w_j}(\boldsymbol{w}) = h(|\boldsymbol{w}|)\delta_{ij} + h'(|\boldsymbol{w}|) \frac{w_i w_j}{|\boldsymbol{w}|}.$$

Contracting with $\boldsymbol{\xi} \in \mathbb{R}^3$ gives $\boldsymbol{\xi}^\mathsf{T} DV(\boldsymbol{w})\boldsymbol{\xi} = h(|\boldsymbol{w}|)|\boldsymbol{\xi}|^2 + \frac{h'(|\boldsymbol{w}|)}{|\boldsymbol{w}|}(\boldsymbol{w} \cdot \boldsymbol{\xi})^2$. Because $h(|\boldsymbol{w}|) \geq 0$ and $h'(|\boldsymbol{w}|) \geq 0$, dropping the non-negative second term establishes Inequality (20). Since the Hessian $D^2\Psi(\boldsymbol{w}) = DV(\boldsymbol{w})$ is positive semi-definite everywhere on $\mathbb{R}^3$, Ψ is convex, and integrating $\frac{\mathrm{d}}{\mathrm{d}\theta} V(\boldsymbol{b} + \theta(\boldsymbol{a} - \boldsymbol{b})) \cdot (\boldsymbol{a} - \boldsymbol{b}) = (\boldsymbol{a} - \boldsymbol{b})^\mathsf{T} DV(\boldsymbol{b} + \theta(\boldsymbol{a} - \boldsymbol{b}))(\boldsymbol{a} - \boldsymbol{b}) \geq 0$ over $\theta \in [0, 1]$ proves the monotonicity inequality (19).

(3) By definition of Φ, $1 - \Phi(|\boldsymbol{w}|/\Omega_c) = 0$ whenever $|\boldsymbol{w}| \geq 2\Omega_c$, and $0 \leq 1 - \Phi(|\boldsymbol{w}|/\Omega_c) \leq 1$ when $|\boldsymbol{w}| < 2\Omega_c$. Consequently, $(1 - \Phi(|\boldsymbol{w}|/\Omega_c))|\boldsymbol{w}|^3 \leq (2\Omega_c)^3 = 8\Omega_c^3$ for all $\boldsymbol{w} \in \mathbb{R}^3$. Rearranging $\Phi(|\boldsymbol{w}|/\Omega_c)|\boldsymbol{w}|^3 = |\boldsymbol{w}|^3 - (1 - \Phi(|\boldsymbol{w}|/\Omega_c))|\boldsymbol{w}|^3 \geq |\boldsymbol{w}|^3 - 8\Omega_c^3$ and multiplying by $C_O \ell^2 > 0$ yields Inequality (21). □

Lemma 3.4 (Integration by Parts of Rotational Dissipation at Energy and Enstrophy Levels)**.** *For any smooth solenoidal velocity field $\boldsymbol{u} \in H^2_\sigma(\mathbb{T}^3)$ with vorticity $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$, the rotational selective dissipation operator $F_{\text{diss}}(\boldsymbol{\omega}) = -\nabla \times V(\boldsymbol{\omega})$ satisfies:*

$$\langle F_{\text{diss}}(\boldsymbol{\omega}), \boldsymbol{u} \rangle = -\int_{\mathbb{T}^3} V(\boldsymbol{\omega}) \cdot \boldsymbol{\omega} \, \mathrm{d}x = -C_O \ell^2 \int_{\mathbb{T}^3} \Phi\left(\frac{|\boldsymbol{\omega}|}{\Omega_c}\right) |\boldsymbol{\omega}|^3 \, \mathrm{d}x \leq 0, \tag{22}$$

$$\langle F_{\text{diss}}(\boldsymbol{\omega}), -\Delta \boldsymbol{u} \rangle = -\int_{\mathbb{T}^3} \partial_k \omega_i \frac{\partial V_i}{\partial \omega_j}(\boldsymbol{\omega}) \partial_k \omega_j \, \mathrm{d}x \leq -C_O \ell^2 \int_{\mathbb{T}^3} \Phi\left(\frac{|\boldsymbol{\omega}|}{\Omega_c}\right) |\boldsymbol{\omega}||\nabla \boldsymbol{\omega}|^2 \, \mathrm{d}x \leq 0. \tag{23}$$

Proof. On the periodic boundaryless torus $\mathbb{T}^3$, vector integration by parts of the curl operator gives $\int_{\mathbb{T}^3} (\nabla \times \boldsymbol{A}) \cdot \boldsymbol{B} \, \mathrm{d}x = \int_{\mathbb{T}^3} \boldsymbol{A} \cdot (\nabla \times \boldsymbol{B}) \, \mathrm{d}x$. Applying this with $\boldsymbol{A} = -V(\boldsymbol{\omega})$ and $\boldsymbol{B} = \boldsymbol{u}$ (where $\nabla \times \boldsymbol{u} = \boldsymbol{\omega}$) immediately proves Equation (22). Next, because $\nabla \cdot \boldsymbol{u} = 0$, the vector Laplacian identity gives $-\Delta \boldsymbol{u} = \nabla \times (\nabla \times \boldsymbol{u}) - \nabla(\nabla \cdot \boldsymbol{u}) = \nabla \times \boldsymbol{\omega}$. Integrating by parts twice yields:

$$\langle -\nabla \times V(\boldsymbol{\omega}), -\Delta \boldsymbol{u} \rangle = -\int_{\mathbb{T}^3} (\nabla \times V(\boldsymbol{\omega})) \cdot (\nabla \times \boldsymbol{\omega}) \, \mathrm{d}x = \int_{\mathbb{T}^3} V(\boldsymbol{\omega}) \cdot \Delta \boldsymbol{\omega} \, \mathrm{d}x = -\sum_{k=1}^{3} \int_{\mathbb{T}^3} \partial_k V_i(\boldsymbol{\omega}) \partial_k \omega_i \, \mathrm{d}x,$$

where we used $\nabla \times (\nabla \times \boldsymbol{\omega}) = -\Delta \boldsymbol{\omega}$ since $\nabla \cdot \boldsymbol{\omega} = 0$. By the chain rule, $\partial_k V_i(\boldsymbol{\omega}) = \frac{\partial V_i}{\partial \omega_j}(\boldsymbol{\omega}) \partial_k \omega_j$. Applying the Jacobian coercivity bound (20) of Lemma 3.3 to each spatial derivative vector $\boldsymbol{\xi}^{(k)} = \partial_k \boldsymbol{\omega}$ gives $\sum_{k=1}^3 (\partial_k \boldsymbol{\omega})^\mathsf{T} DV(\boldsymbol{\omega})(\partial_k \boldsymbol{\omega}) \geq h(|\boldsymbol{\omega}|) \sum_{k=1}^3 |\partial_k \boldsymbol{\omega}|^2 = C_O \ell^2 \Phi(|\boldsymbol{\omega}|/\Omega_c)|\boldsymbol{\omega}||\nabla \boldsymbol{\omega}|^2$, establishing Inequality (23). □

3.3 Periodic Calderón–Zygmund Singular Integral Estimates on $\mathbb{T}^3$

Lemma 3.5 (Calderón–Zygmund Gradient Bounds and Cubic Vortex Stretching Estimate on $\mathbb{T}^3$)**.** *For every $1 < p < \infty$, there exists an explicit analytical constant $C_p > 0$ depending only on p such that for all solenoidal velocity fields $\boldsymbol{u} \in W^{1,p}(\mathbb{T}^3) \cap L^2_\sigma(\mathbb{T}^3)$ with vorticity $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$:*

$$\|\nabla \boldsymbol{u}\|_{L^p(\mathbb{T}^3)} \leq C_p \|\boldsymbol{\omega}\|_{L^p(\mathbb{T}^3)}, \qquad \|S(\boldsymbol{u})\|_{L^p(\mathbb{T}^3)} \leq C_p \|\boldsymbol{\omega}\|_{L^p(\mathbb{T}^3)}. \tag{24}$$

In particular, for $p = 2$, $\|\nabla \boldsymbol{u}\|_{L^2} = \|\boldsymbol{\omega}\|_{L^2}$ holds with exact constant $C_2 = 1$, and for $p = 3$, the cubic vortex stretching integral satisfies:

$$\left| \int_{\mathbb{T}^3} (\boldsymbol{\omega} \cdot \nabla)\boldsymbol{u} \cdot \boldsymbol{\omega} \, \mathrm{d}x \right| = \left| \int_{\mathbb{T}^3} \boldsymbol{\omega} \cdot S(\boldsymbol{u})\boldsymbol{\omega} \, \mathrm{d}x \right| \leq \|S(\boldsymbol{u})\|_{L^3(\mathbb{T}^3)} \|\boldsymbol{\omega}\|^2_{L^3(\mathbb{T}^3)} \leq C_3 \|\boldsymbol{\omega}\|^3_{L^3(\mathbb{T}^3)}. \tag{25}$$

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Proof. Since $\nabla \cdot \boldsymbol{u} = 0$ and $\int_{\mathbb{T}^3} \boldsymbol{u}\,dx = \mathbf{0}$, the velocity field is recovered from the vorticity via the periodic Biot–Savart law $\boldsymbol{u} = (-\Delta)^{-1}(\nabla \times \boldsymbol{\omega})$. In Fourier space on $\mathbb{T}^3$, for every $\boldsymbol{k} \in \mathbb{Z}^3 \setminus \{\mathbf{0}\}$, the velocity gradient components $(\widehat{\partial_j u_i})_{\boldsymbol{k}} = ik_j \hat{u}_{i,\boldsymbol{k}}$ are expressed in terms of $\hat{\boldsymbol{\omega}}_{\boldsymbol{k}} = i\boldsymbol{k} \times \hat{\boldsymbol{u}}_{\boldsymbol{k}}$ as:

$$(\widehat{\partial_j u_i})_{\boldsymbol{k}} = -\sum_{l,m=1}^{3} \frac{k_j k_l}{|\boldsymbol{k}|^2} \varepsilon_{ilm} \hat{\omega}_{m,\boldsymbol{k}} = \sum_{l,m=1}^{3} (\widehat{\mathcal{R}_j \mathcal{R}_l})_{\boldsymbol{k}} \varepsilon_{ilm} \hat{\omega}_{m,\boldsymbol{k}},$$

where $\widehat{(\mathcal{R}_j f)}_{\boldsymbol{k}} := \frac{ik_j}{|\boldsymbol{k}|} \hat{f}_{\boldsymbol{k}}$ denotes the j-th periodic Riesz transform of order zero on $\mathbb{T}^3$. By the classical Calderón–Zygmund singular integral theory and transference of the Marcinkiewicz multiplier theorem to the torus $\mathbb{T}^3$ [15, 21], $\mathcal{R}_j \mathcal{R}_l$ is a bounded linear operator from $L^p(\mathbb{T}^3)$ to $L^p(\mathbb{T}^3)$ for every $1 < p < \infty$, which proves Estimate (24). Finally, since the skew-symmetric spin tensor satisfies $W(\boldsymbol{u})\boldsymbol{\omega} = -\frac{1}{2}\boldsymbol{\omega} \times \boldsymbol{\omega} = \mathbf{0}$ pointwise, we have $(\boldsymbol{\omega} \cdot \nabla)\boldsymbol{u} \cdot \boldsymbol{\omega} = \boldsymbol{\omega} \cdot S(\boldsymbol{u})\boldsymbol{\omega}$. Applying Hölder's inequality with exponents $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and Estimate (24) at $p = 3$ establishes Inequality (25). □

4 Global Strong Well-Posedness, Higher Sobolev Regularity, and Vanishing Filter Limit ($\ell \to 0^+$)

We now synthesize the structural estimates of Section 3 to prove the three main analytical theorems of Part 3: (i) unconditional global existence and enstrophy boundedness of strong solutions for arbitrary initial data $\boldsymbol{u}_0 \in H^1_\sigma(\mathbb{T}^3)$ at any fixed $\ell > 0$ (Theorem 4.1); (ii) unconditional global uniqueness and continuous dependence on initial data (Theorem 4.2); and (iii) convergence to classical Leray–Hopf weak solutions in the vanishing filter limit $\ell \to 0^+$ (Theorem 4.3).

4.1 Global Existence and Unconditional Enstrophy Closure of Strong Solutions

For any initial velocity $\boldsymbol{u}_0 \in H^1_\sigma(\mathbb{T}^3)$ and external solenoidal forcing $\boldsymbol{f} \in L^2_{\mathrm{loc}}([0,\infty); L^2_\sigma(\mathbb{T}^3))$, we define the deterministic kinetic velocity envelope $M(t)$ and the total energy functional bound $\mathcal{E}(t)$ for $t \geq 0$ by:

$$M(t) := \|\boldsymbol{u}_0\| + \int_0^t \|\boldsymbol{f}(s)\|\,ds, \qquad \mathcal{E}(t) := \frac{1}{2}\|\boldsymbol{u}_0\|^2 + M(t)\int_0^t \|\boldsymbol{f}(s)\|\,ds \leq M(t)^2. \tag{26}$$

Theorem 4.1 (Unconditional Global Existence and Strong Regularity of $\mathrm{NS}_{\mathrm{TZT}}$)**.** *Let $\ell > 0$, $\nu > 0$, $C_O > 0$, $c_X \in \mathbb{R}$, $\Omega_c > 0$, and $\varepsilon > 0$ be fixed physical parameters. For any arbitrarily large initial data $\boldsymbol{u}_0 \in H^1_\sigma(\mathbb{T}^3)$ and external force $\boldsymbol{f} \in L^2_{\mathrm{loc}}([0,\infty); L^2_\sigma(\mathbb{T}^3))$, there exists a global strong solution $\boldsymbol{u}$ to the $\mathrm{NS}_{\mathrm{TZT}}$ system* (12) *on $[0,\infty) \times \mathbb{T}^3$ satisfying, for every finite horizon $T > 0$:*

$$\begin{gathered} \boldsymbol{u} \in C([0,T]; H^1_\sigma(\mathbb{T}^3)) \cap L^2(0,T; H^2_\sigma(\mathbb{T}^3)), \qquad \boldsymbol{\omega} \in L^3((0,T) \times \mathbb{T}^3), \\ \partial_t \boldsymbol{u} \in L^{3/2}((0,T) \times \mathbb{T}^3) \cap L^{4/3}\big(0,T; (H^1_\sigma(\mathbb{T}^3))^*\big). \end{gathered} \tag{27}$$

Specifically, for all $t \geq 0$, the global enstrophy $\|\boldsymbol{\omega}(t)\|^2 = \|\nabla \boldsymbol{u}(t)\|^2$ obeys the explicit, Grönwall-free upper bound:

$$\begin{aligned} &\|\boldsymbol{\omega}(t)\|^2 + \nu \int_0^t \|\nabla \boldsymbol{\omega}(s)\|^2\,ds + 2C_O\ell^2 \int_0^t \int_{\mathbb{T}^3} \Phi\left(\frac{|\boldsymbol{\omega}|}{\Omega_c}\right) |\boldsymbol{\omega}||\nabla \boldsymbol{\omega}|^2\,dx\,ds \\ &\qquad \leq \|\boldsymbol{\omega}_0\|^2 + 2C_3\left(\frac{\mathcal{E}(t)}{C_O\ell^2} + 8\Omega_c^3(2\pi)^3 t\right) + \frac{1}{\nu}\int_0^t \left(|c_X|\Omega_c M(s) + \|\boldsymbol{f}(s)\|\right)^2 ds < \infty, \end{aligned} \tag{28}$$

where $\boldsymbol{\omega}_0 = \nabla \times \boldsymbol{u}_0$, guaranteeing global regularity without finite-time vorticity blow-up. Furthermore, if $\boldsymbol{u}_0 \in H^m_\sigma(\mathbb{T}^3)$ and $\boldsymbol{f} \in L^2_{\mathrm{loc}}([0,\infty); H^{m-1}_\sigma(\mathbb{T}^3))$ for any integer $m \geq 2$, then $\boldsymbol{u} \in L^\infty_{\mathrm{loc}}([0,\infty); H^m_\sigma(\mathbb{T}^3)) \cap L^2_{\mathrm{loc}}([0,\infty); H^{m+1}_\sigma(\mathbb{T}^3))$.

Proof. We establish Theorem 4.1 in four rigorous analytical steps via spectral Galerkin approximations, energy-level L^3 vorticity absorption, Grönwall-free enstrophy closure, and Minty–Browder monotone compactness.

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Step 1: Spectral Galerkin Approximation and Exact Kinetic Energy Identity. Let $\{w_k\}_{k=1}^{\infty} \subset C^\infty(\mathbb{T}^3) \cap L^2_\sigma(\mathbb{T}^3)$ be the orthonormal eigenbasis of the Stokes operator $A := -\mathbb{P}\Delta = -\Delta$ on $L^2_\sigma(\mathbb{T}^3)$ ordered by non-decreasing eigenvalues $1 = \lambda_1 \le \lambda_2 \le \cdots \to \infty$, and let $\mathbb{P}_N : L^2(\mathbb{T}^3) \to \mathcal{V}_N := \operatorname{span}\{w_1, \dots, w_N\}$ denote the finite-dimensional Fourier–Galerkin orthogonal projection, which commutes with $\mathbb{P}$, ∇, $\nabla\times$, and Δ. Consider the finite-dimensional ODE system on $\mathcal{V}_N$ for $u_N(t) = \sum_{k=1}^N c_{N,k}(t) w_k$ with $\omega_N = \nabla \times u_N$:

$$\begin{aligned} \partial_t u_N + \mathbb{P}_N\big[(u_N \cdot \nabla) u_N\big] &= \nu \Delta u_N - \mathbb{P}_N\big(\nabla \times V(\omega_N)\big) + c_X \Omega_c \mathbb{P}_N G(u_N, \omega_N) + \mathbb{P}_N f, \\ u_N(0) &= \mathbb{P}_N u_0. \end{aligned} \tag{29}$$

By the Cauchy–Peano–Picard theorem, the smooth finite-dimensional system (29) admits a local maximal solution on $[0, T_N)$. Taking the $L^2(\mathbb{T}^3)$ inner product of Equation (29) with $u_N \in \mathcal{V}_N$, the convective term vanishes by incompressibility ($\langle (u_N \cdot \nabla) u_N, u_N \rangle = -\frac{1}{2} \int_{\mathbb{T}^3} (\nabla \cdot u_N) |u_N|^2 \, dx = 0$), and the gyroscopic alignment term vanishes identically by Lemma 3.1 ($\langle \mathbb{P}_N G(u_N, \omega_N), u_N \rangle = \langle G(u_N, \omega_N), u_N \rangle = 0$). Applying Equation (22) of Lemma 3.4 to the rotational dissipation term gives the exact differential energy balance:

$$\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|u_N(t)\|^2 + \nu \|\nabla u_N(t)\|^2 + C_O \ell^2 \int_{\mathbb{T}^3} \Phi\left(\frac{|\omega_N|}{\Omega_c} \right) |\omega_N|^3 \, dx = \langle f(t), u_N(t) \rangle \le \|f(t)\| \|u_N(t)\|. \tag{30}$$

Dropping the non-negative viscous and TZT dissipation terms yields $\frac{\mathrm{d}}{\mathrm{d}t} \|u_N(t)\| \le \|f(t)\|$, whence $\|u_N(t)\| \le \|u_0\| + \int_0^t \|f(s)\| \, ds = M(t)$ for all $t \ge 0$ (which immediately guarantees $T_N = \infty$). Integrating Equation (30) over $[0, t]$ yields the uniform kinetic energy inequality:

$$\frac{1}{2} \|u_N(t)\|^2 + \nu \int_0^t \|\nabla u_N(s)\|^2 \, ds + C_O \ell^2 \int_0^t \int_{\mathbb{T}^3} \Phi\left(\frac{|\omega_N|}{\Omega_c} \right) |\omega_N|^3 \, dx \, ds \le \mathcal{E}(t). \tag{31}$$

Step 2: Unconditional Space-Time L^3 Vorticity Bound. By the pointwise cubic coercivity bound (21) of Lemma 3.3, $|\omega_N(x,s)|^3 \le \Phi(|\omega_N(x,s)|/\Omega_c) |\omega_N(x,s)|^3 + 8\Omega_c^3$ holds everywhere on $\mathbb{T}^3 \times [0, t]$. Integrating this pointwise bound over $[0, t] \times \mathbb{T}^3$ and substituting the third term on the left-hand side of Inequality (31) yields the uniform space-time L^3 vorticity estimate:

$$\int_0^t \|\omega_N(s)\|_{L^3(\mathbb{T}^3)}^3 \, ds = \int_0^t \int_{\mathbb{T}^3} |\omega_N(x,s)|^3 \, dx \, ds \le \frac{\mathcal{E}(t)}{C_O \ell^2} + 8\Omega_c^3 |\mathbb{T}^3| t = \frac{\mathcal{E}(t)}{C_O \ell^2} + 8\Omega_c^3 (2\pi)^3 t < \infty. \tag{32}$$

Step 3: Grönwall-Free Enstrophy Closure in $H^1_\sigma(\mathbb{T}^3)$. Because $\mathcal{V}_N$ is invariant under $-\Delta$, we may test the Galerkin equation (29) in $L^2(\mathbb{T}^3)$ with $-\Delta u_N = \nabla \times \omega_N \in \mathcal{V}_N$. Integrating by parts gives $\langle \partial_t u_N, -\Delta u_N \rangle = \frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|\nabla u_N\|^2 = \frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|\omega_N\|^2$ and $\langle \nu \Delta u_N, -\Delta u_N \rangle = -\nu \|\Delta u_N\|^2 = -\nu \|\nabla \omega_N\|^2$. For the convective term, standard curl integration by parts yields $\langle -(u_N \cdot \nabla) u_N, -\Delta u_N \rangle = \int_{\mathbb{T}^3} \omega_N \cdot S(u_N) \omega_N \, dx$. Combining these identities with Inequality (23) of Lemma 3.4 yields the exact enstrophy evolution equation:

$$\begin{aligned} &\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|\omega_N\|^2 + \nu \|\nabla \omega_N\|^2 + C_O \ell^2 \int_{\mathbb{T}^3} \Phi\left(\frac{|\omega_N|}{\Omega_c} \right) |\omega_N| |\nabla \omega_N|^2 \, dx \\ &\qquad \le \int_{\mathbb{T}^3} \omega_N \cdot S(u_N) \omega_N \, dx + \int_{\mathbb{T}^3} \big(c_X \Omega_c G(u_N, \omega_N) + f \big) \cdot (-\Delta u_N) \, dx. \end{aligned} \tag{33}$$

By the Calderón–Zygmund estimate (25) of Lemma 3.5, the cubic vortex stretching term is bounded by:

$$\left| \int_{\mathbb{T}^3} \omega_N \cdot S(u_N) \omega_N \, dx \right| \le C_3 \|\omega_N\|_{L^3(\mathbb{T}^3)}^3. \tag{34}$$

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For the gyroscopic and external forcing terms, applying the Cauchy–Schwarz inequality, $\|\Delta \boldsymbol{u}_N\| = \|\nabla \boldsymbol{\omega}_N\|$, Young's inequality ($ab \le \frac{\nu}{2}a^2 + \frac{1}{2\nu}b^2$), and Estimate (16) ($\|G(\boldsymbol{u}_N, \boldsymbol{\omega}_N)\| \le \|\boldsymbol{u}_N\| \le M(t)$) yields:

$$\begin{aligned}\left| \int_{\mathbb{T}^3} \big(c_X \Omega_c G(\boldsymbol{u}_N, \boldsymbol{\omega}_N) + \boldsymbol{f}\big) \cdot (-\Delta \boldsymbol{u}_N)\, \mathrm{d}x \right| &\le \big(|c_X|\Omega_c \|\boldsymbol{u}_N\| + \|\boldsymbol{f}\|\big)\, \|\nabla \boldsymbol{\omega}_N\| \\ &\le \frac{\nu}{2}\|\nabla \boldsymbol{\omega}_N\|^2 + \frac{1}{2\nu}\left(|c_X|\Omega_c M(t) + \|\boldsymbol{f}(t)\|\right)^2 .\end{aligned} \tag{35}$$

Substituting Estimates (34) and (35) into Inequality (33), absorbing $\frac{\nu}{2}\|\nabla \boldsymbol{\omega}_N\|^2$ into the left-hand side, multiplying by 2, and integrating over $[0, t]$ gives:

$$\begin{aligned}\|\boldsymbol{\omega}_N(t)\|^2 + \nu \int_0^t \|\nabla \boldsymbol{\omega}_N(s)\|^2\, \mathrm{d}s + 2C_O \ell^2 \int_0^t \int_{\mathbb{T}^3} \Phi\left(\frac{|\boldsymbol{\omega}_N|}{\Omega_c}\right) |\boldsymbol{\omega}_N| |\nabla \boldsymbol{\omega}_N|^2\, \mathrm{d}x\, \mathrm{d}s \\ \le \|\boldsymbol{\omega}_0\|^2 + 2C_3 \int_0^t \|\boldsymbol{\omega}_N(s)\|_{L^3}^3\, \mathrm{d}s + \frac{1}{\nu}\int_0^t \left(|c_X|\Omega_c M(s) + \|\boldsymbol{f}(s)\|\right)^2 \mathrm{d}s.\end{aligned}$$

Crucially, because $\int_0^t \|\boldsymbol{\omega}_N(s)\|_{L^3}^3\, \mathrm{d}s$ is already bounded a priori by Equation (32) from the kinetic energy balance, substituting Equation (32) directly yields Estimate (28) for $\boldsymbol{u}_N$ without any Grönwall iteration.

Step 4: Compactness, Monotone Limit Passage, and Higher Regularity. By Inequality (28), on every finite time interval $[0, T]$, the sequence $\{\boldsymbol{u}_N\}$ is uniformly bounded in $L^\infty(0, T; H^1_\sigma(\mathbb{T}^3)) \cap L^2(0, T; H^2_\sigma(\mathbb{T}^3))$, and $\{\boldsymbol{\omega}_N\}$ is uniformly bounded in $L^\infty(0, T; L^2) \cap L^2(0, T; H^1) \cap L^{10/3}((0, T) \times \mathbb{T}^3)$. Moreover, since $|V(\boldsymbol{\omega}_N)| \le C_O \ell^2 |\boldsymbol{\omega}_N|^2$ and $\|\boldsymbol{\omega}_N\|^2_{L^4(\mathbb{T}^3)} \le C\|\boldsymbol{\omega}_N\|_{L^2}^{1/2}\|\nabla \boldsymbol{\omega}_N\|_{L^2}^{3/2}$, $V(\boldsymbol{\omega}_N)$ is uniformly bounded in $L^{4/3}(0, T; L^2(\mathbb{T}^3)) \cap L^{5/3}((0, T) \times \mathbb{T}^3)$, and by the weighted enstrophy dissipation bound $|\boldsymbol{\omega}_N|^{1/2}\nabla \boldsymbol{\omega}_N \in L^2((0, T) \times \mathbb{T}^3)$ on the support of Φ, Hölder's inequality yields $|\nabla V(\boldsymbol{\omega}_N)| \le C(|\boldsymbol{\omega}_N|^{1/2} + \Omega_c^{1/2})(|\boldsymbol{\omega}_N|^{1/2}|\nabla \boldsymbol{\omega}_N| + \Omega_c^{1/2}|\nabla \boldsymbol{\omega}_N|) \in L^{3/2}((0, T) \times \mathbb{T}^3)$. Consequently, $\{\nabla \times V(\boldsymbol{\omega}_N)\}$ and $\{\partial_t \boldsymbol{u}_N\}$ are uniformly bounded in $L^{3/2}((0, T) \times \mathbb{T}^3) \cap L^{4/3}(0, T; (H^1_\sigma(\mathbb{T}^3))^*)$. By the Aubin–Lions–Simon compactness theorem [19], there exists a subsequence converging strongly:

$$\begin{aligned}&\boldsymbol{u}_N \to \boldsymbol{u} \quad \text{strongly in } C\big([0, T]; L^2_\sigma(\mathbb{T}^3)\big) \cap L^2\big(0, T; H^1_\sigma(\mathbb{T}^3)\big), \\ &\boldsymbol{\omega}_N \to \boldsymbol{\omega} \quad \text{strongly in } L^2\big((0, T) \times \mathbb{T}^3\big) \text{ and a.e.}\end{aligned}$$

Because $\boldsymbol{\omega}_N \to \boldsymbol{\omega}$ strongly in $L^2((0, T) \times \mathbb{T}^3)$ and almost everywhere, and $\{\boldsymbol{\omega}_N\}$ is bounded in $L^\infty(0, T; L^2) \cap L^2(0, T; L^6) \hookrightarrow L^{10/3}((0, T) \times \mathbb{T}^3)$, Vitali's convergence theorem implies $\boldsymbol{\omega}_N \to \boldsymbol{\omega}$ strongly in $L^3((0, T) \times \mathbb{T}^3)$ and $V(\boldsymbol{\omega}_N) \to V(\boldsymbol{\omega})$ strongly in $L^{5/3}((0, T) \times \mathbb{T}^3)$ (and also via the standard Minty–Browder monotonicity trick using Lemma 3.3(1)). Similarly, by Lemma 3.2, $G(\boldsymbol{u}_N, \boldsymbol{\omega}_N) \to G(\boldsymbol{u}, \boldsymbol{\omega})$ strongly in $L^2((0, T) \times \mathbb{T}^3)$. Passing to the limit in Equation (29) shows that $\boldsymbol{u}$ satisfies Equation (13) in $L^{3/2}((0, T) \times \mathbb{T}^3)$ (and in $L^2(0, T; L^2_\sigma)$ whenever $\boldsymbol{u}_0 \in H^2_\sigma(\mathbb{T}^3)$). Weak lower semicontinuity of norms yields Estimate (28), and bootstrap differentiation (detailed in Appendix A) establishes higher $H^m_\sigma(\mathbb{T}^3)$ regularity. □

4.2 Unconditional Uniqueness and Continuous Dependence in $H^1_\sigma(\mathbb{T}^3)$

Theorem 4.2 (Unconditional Global Uniqueness and Continuous Dependence)**.** *Let $\boldsymbol{u}_1, \boldsymbol{u}_2 \in C([0, T]; H^1_\sigma(\mathbb{T}^3)) \cap L^2(0, T; H^2_\sigma(\mathbb{T}^3))$ be two strong solutions to the* $\mathrm{NS}_{\mathrm{TZT}}$ *system* (12) *on $[0, T]$ corresponding to initial data $\boldsymbol{u}_{1,0}, \boldsymbol{u}_{2,0} \in H^1_\sigma(\mathbb{T}^3)$ and the same forcing $\boldsymbol{f}$. Then the difference $\boldsymbol{w}(t) := \boldsymbol{u}_1(t) - \boldsymbol{u}_2(t)$ satisfies the stability estimate:*

$$\|\boldsymbol{w}(t)\|^2 + \nu \int_0^t \|\nabla \boldsymbol{w}(s)\|^2\, \mathrm{d}s \le \|\boldsymbol{u}_{1,0} - \boldsymbol{u}_{2,0}\|^2 \exp\left(\int_0^t \mathcal{K}_{\mathrm{uniq}}(s)\, \mathrm{d}s\right), \qquad \forall\, t \in [0, T], \tag{36}$$

where $\mathcal{K}_{\mathrm{uniq}}(s) := C_L \nu^{-3}\|\nabla \boldsymbol{u}_2(s)\|^4 + 2\nu^{-1}c_X^2 \Omega_c^2 L^2_{\beta,\varepsilon}\|\boldsymbol{u}_1(s)\|^2_{L^\infty(\mathbb{T}^3)} \in L^1(0, T)$. In particular, if $\boldsymbol{u}_{1,0} = \boldsymbol{u}_{2,0}$, then $\boldsymbol{u}_1(t) \equiv \boldsymbol{u}_2(t)$ on $[0, T]$.

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Proof. Subtracting the projected evolution equations (13) for $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$, the difference $\boldsymbol{w} = \boldsymbol{u}_1 - \boldsymbol{u}_2$ with vorticity difference $\boldsymbol{\zeta} := \nabla \times \boldsymbol{w} = \boldsymbol{\omega}_1 - \boldsymbol{\omega}_2$ satisfies:

$$\partial_t \boldsymbol{w} + \mathbb{P}\big[(\boldsymbol{u}_1 \cdot \nabla)\boldsymbol{w} + (\boldsymbol{w} \cdot \nabla)\boldsymbol{u}_2\big] = \nu \Delta \boldsymbol{w} - \nabla \times \big(V(\boldsymbol{\omega}_1) - V(\boldsymbol{\omega}_2)\big) + c_X \Omega_c \mathbb{P}\big(G(\boldsymbol{u}_1, \boldsymbol{\omega}_1) - G(\boldsymbol{u}_2, \boldsymbol{\omega}_2)\big).$$

Taking the $L^2(\mathbb{T}^3)$ inner product with $\boldsymbol{w} \in L^2_\sigma(\mathbb{T}^3)$, we note first that $\langle (\boldsymbol{u}_1 \cdot \nabla)\boldsymbol{w}, \boldsymbol{w} \rangle = 0$ since $\nabla \cdot \boldsymbol{u}_1 = 0$. Second, integrating by parts the rotational dissipation difference and invoking the convexity/monotonicity inequality (19) of Lemma 3.3 gives:

$$-\langle \nabla \times (V(\boldsymbol{\omega}_1) - V(\boldsymbol{\omega}_2)), \boldsymbol{w} \rangle = -\int_{\mathbb{T}^3} \big(V(\boldsymbol{\omega}_1) - V(\boldsymbol{\omega}_2)\big) \cdot (\boldsymbol{\omega}_1 - \boldsymbol{\omega}_2)\, \mathrm{d}x \le 0. \tag{37}$$

Third, for the gyroscopic difference term, recall from Lemma 3.2 that $G(\boldsymbol{u}_1, \boldsymbol{\omega}_1) - G(\boldsymbol{u}_2, \boldsymbol{\omega}_2) = \boldsymbol{w} \times \beta(\boldsymbol{\omega}_2) + \boldsymbol{u}_1 \times (\beta(\boldsymbol{\omega}_1) - \beta(\boldsymbol{\omega}_2))$. Taking the dot product with $\boldsymbol{w}$, the first piece vanishes identically since $(\boldsymbol{w} \times \beta(\boldsymbol{\omega}_2)) \cdot \boldsymbol{w} \equiv 0$ pointwise. For the remaining piece, applying Inequality (17), Hölder's inequality, $\|\boldsymbol{\zeta}\| = \|\nabla \boldsymbol{w}\|$, and Young's inequality yields:

$$\begin{aligned} c_X \Omega_c \langle G(\boldsymbol{u}_1, \boldsymbol{\omega}_1) - G(\boldsymbol{u}_2, \boldsymbol{\omega}_2), \boldsymbol{w} \rangle &= c_X \Omega_c \int_{\mathbb{T}^3} \big(\boldsymbol{u}_1 \times (\beta(\boldsymbol{\omega}_1) - \beta(\boldsymbol{\omega}_2))\big) \cdot \boldsymbol{w}\, \mathrm{d}x \\ &\le |c_X| \Omega_c L_{\beta,\varepsilon} \|\boldsymbol{u}_1\|_{L^\infty(\mathbb{T}^3)} \|\boldsymbol{\omega}_1 - \boldsymbol{\omega}_2\| \|\boldsymbol{w}\| \\ &\le \frac{\nu}{4} \|\nabla \boldsymbol{w}\|^2 + \frac{c_X^2 \Omega_c^2 L_{\beta,\varepsilon}^2}{\nu} \|\boldsymbol{u}_1\|^2_{L^\infty(\mathbb{T}^3)} \|\boldsymbol{w}\|^2. \end{aligned} \tag{38}$$

Fourth, for the convective difference term $-\langle (\boldsymbol{w} \cdot \nabla)\boldsymbol{u}_2, \boldsymbol{w} \rangle$, applying Hölder's inequality, Ladyzhenskaya's 3D interpolation inequality $\|\boldsymbol{w}\|^2_{L^4(\mathbb{T}^3)} \le C_{\mathrm{Lad}} \|\boldsymbol{w}\|^{1/2} \|\nabla \boldsymbol{w}\|^{3/2}$ [11], and Young's inequality with exponents $(4, \frac{4}{3})$ gives:

$$|\langle (\boldsymbol{w} \cdot \nabla)\boldsymbol{u}_2, \boldsymbol{w} \rangle| \le \|\nabla \boldsymbol{u}_2\| \|\boldsymbol{w}\|^2_{L^4} \le C_{\mathrm{Lad}} \|\nabla \boldsymbol{u}_2\| \|\boldsymbol{w}\|^{1/2} \|\nabla \boldsymbol{w}\|^{3/2} \le \frac{\nu}{4} \|\nabla \boldsymbol{w}\|^2 + \frac{C_L}{2\nu^3} \|\nabla \boldsymbol{u}_2\|^4 \|\boldsymbol{w}\|^2. \tag{39}$$

Combining Equations (37)–(39) yields $\frac{\mathrm{d}}{\mathrm{d}t} \|\boldsymbol{w}(t)\|^2 + \nu \|\nabla \boldsymbol{w}(t)\|^2 \le \mathcal{K}_{\mathrm{uniq}}(t) \|\boldsymbol{w}(t)\|^2$. Since $\boldsymbol{u}_2 \in L^\infty(0, T; H^1_\sigma)$ implies $\|\nabla \boldsymbol{u}_2\|^4 \in L^\infty(0, T) \subset L^1(0, T)$, and Agmon's 3D inequality $\|\boldsymbol{u}_1\|^2_{L^\infty} \le C_{\mathrm{Ag}} \|\boldsymbol{u}_1\|_{H^1} \|\boldsymbol{u}_1\|_{H^2} \in L^1(0, T)$ since $\boldsymbol{u}_1 \in L^\infty H^1 \cap L^2 H^2$, the coefficient $\mathcal{K}_{\mathrm{uniq}}(t)$ is integrable on $[0, T]$. Applying Grönwall's lemma completes the proof. □

4.3 Vanishing Filter Scale Limit ($\ell \to 0^+$) and Convergence to Leray–Hopf Weak Solutions

Theorem 4.3 (Vanishing Filter Scale Limit $\ell \to 0^+$ and Leray–Hopf Weak Convergence)**.** *Let $\{\ell_k\}_{k=1}^\infty$ be any sequence of positive filter scales converging to zero ($\ell_k \to 0^+$) with $c_X = \tilde{c}_X \ell_k$ (or fixed $c_X = 0$), and let $\boldsymbol{u}^{(\ell_k)}$ denote the unique global strong solution to the* $\mathrm{NS}_{\mathrm{TZT}}$ *system* (12) *with initial data $\boldsymbol{u}_0 \in H^1_\sigma(\mathbb{T}^3)$ and forcing $\boldsymbol{f} \in L^2_{\mathrm{loc}}([0, \infty); L^2_\sigma(\mathbb{T}^3))$. Then for every $T > 0$, there exists a subsequence (still denoted $\boldsymbol{u}^{(\ell_k)}$) and a limit velocity field $\boldsymbol{u} \in L^\infty(0, T; L^2_\sigma(\mathbb{T}^3)) \cap L^2(0, T; H^1_\sigma(\mathbb{T}^3))$ such that as $\ell_k \to 0^+$:*

$$\begin{gathered} \boldsymbol{u}^{(\ell_k)} \overset{*}{\rightharpoonup} \boldsymbol{u} \text{ weakly-* in } L^\infty(0, T; L^2_\sigma), \qquad \boldsymbol{u}^{(\ell_k)} \rightharpoonup \boldsymbol{u} \text{ weakly in } L^2(0, T; H^1_\sigma), \\ \boldsymbol{u}^{(\ell_k)} \to \boldsymbol{u} \text{ strongly in } L^2(0, T; L^2_\sigma), \end{gathered} \tag{40}$$

and the limit $\boldsymbol{u}$ is a global Leray–Hopf weak solution of the classical three-dimensional incompressible Navier–Stokes equations ($\partial_t \boldsymbol{u} + \mathbb{P}[(\boldsymbol{u} \cdot \nabla)\boldsymbol{u}] = \nu \Delta \boldsymbol{u} + \boldsymbol{f}$) satisfying the classical energy inequality $\frac{1}{2}\|\boldsymbol{u}(t)\|^2 + \nu \int_0^t \|\nabla \boldsymbol{u}(s)\|^2\, \mathrm{d}s \le \frac{1}{2}\|\boldsymbol{u}_0\|^2 + \int_0^t \langle \boldsymbol{f}(s), \boldsymbol{u}(s) \rangle\, \mathrm{d}s$ for almost every $t \ge 0$.

Proof. By the exact kinetic energy inequality (31) established in Step 1 of Theorem 4.1, the upper bound $\mathcal{E}(T)$ is completely independent of $\ell_k > 0$:

$$\sup_{t \in [0,T]} \frac{1}{2} \|\boldsymbol{u}^{(\ell_k)}(t)\|^2 + \nu \int_0^T \|\nabla \boldsymbol{u}^{(\ell_k)}(s)\|^2\, \mathrm{d}s + C_O \ell_k^2 \int_0^T \int_{\mathbb{T}^3} \Phi\left(\frac{|\boldsymbol{\omega}^{(\ell_k)}|}{\Omega_c}\right) |\boldsymbol{\omega}^{(\ell_k)}|^3\, \mathrm{d}x\, \mathrm{d}s \le \mathcal{E}(T). \tag{41}$$

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From the third term of Inequality (41) and $\Phi \in [0,1]$ (so $\Phi \le \Phi^{2/3}$), for any smooth solenoidal test function $\boldsymbol{\varphi} \in C_c^\infty((0,T)\times\mathbb{T}^3;\mathbb{R}^3)$ with $\nabla\cdot\boldsymbol{\varphi}=0$, Hölder's inequality with exponents $(\frac{3}{2},3)$ yields:

$$\begin{aligned}\left|\int_0^T \langle -\nabla\times V(\boldsymbol{\omega}^{(\ell_k)}),\boldsymbol{\varphi}\rangle\,\mathrm{d}t\right| &= \left|\int_0^T\int_{\mathbb{T}^3} C_O\ell_k^2\Phi|\boldsymbol{\omega}^{(\ell_k)}|\boldsymbol{\omega}^{(\ell_k)}\cdot(\nabla\times\boldsymbol{\varphi})\,\mathrm{d}x\,\mathrm{d}t\right| \\ &\le C_O^{1/3}\ell_k^{2/3}\left(C_O\ell_k^2\int_0^T\int_{\mathbb{T}^3}\Phi|\boldsymbol{\omega}^{(\ell_k)}|^3\,\mathrm{d}x\,\mathrm{d}t\right)^{2/3}\|\nabla\times\boldsymbol{\varphi}\|_{L^3} \\ &\le C_O^{1/3}\mathcal{E}(T)^{2/3}\|\nabla\times\boldsymbol{\varphi}\|_{L^3}\,\ell_k^{2/3}\xrightarrow{\ell_k\to0^+}0.\end{aligned}$$

Similarly, $\|F_{\text{axis}}^{(\ell_k)}\|_{L^2} \le |\tilde{c}_X|\ell_k\Omega_c M(T) \to 0$ as $\ell_k\to0^+$. Because $\{\partial_t\boldsymbol{u}^{(\ell_k)}\}$ is uniformly bounded in $L^{4/3}(0,T;(H^2_\sigma(\mathbb{T}^3))^*)$, the Aubin–Lions lemma gives strong convergence $\boldsymbol{u}^{(\ell_k)}\to\boldsymbol{u}$ in $L^2(0,T;L^2_\sigma(\mathbb{T}^3))$, which allows passage to the limit in the quadratic convective term $\int_0^T\langle(\boldsymbol{u}^{(\ell_k)}\cdot\nabla)\boldsymbol{u}^{(\ell_k)},\boldsymbol{\varphi}\rangle\,\mathrm{d}t \to \int_0^T\langle(\boldsymbol{u}\cdot\nabla)\boldsymbol{u},\boldsymbol{\varphi}\rangle\,\mathrm{d}t$. Dropping the non-negative ℓ_k^2 dissipation term in Inequality (41) and applying weak lower semicontinuity yields the classical Leray–Hopf energy inequality. □

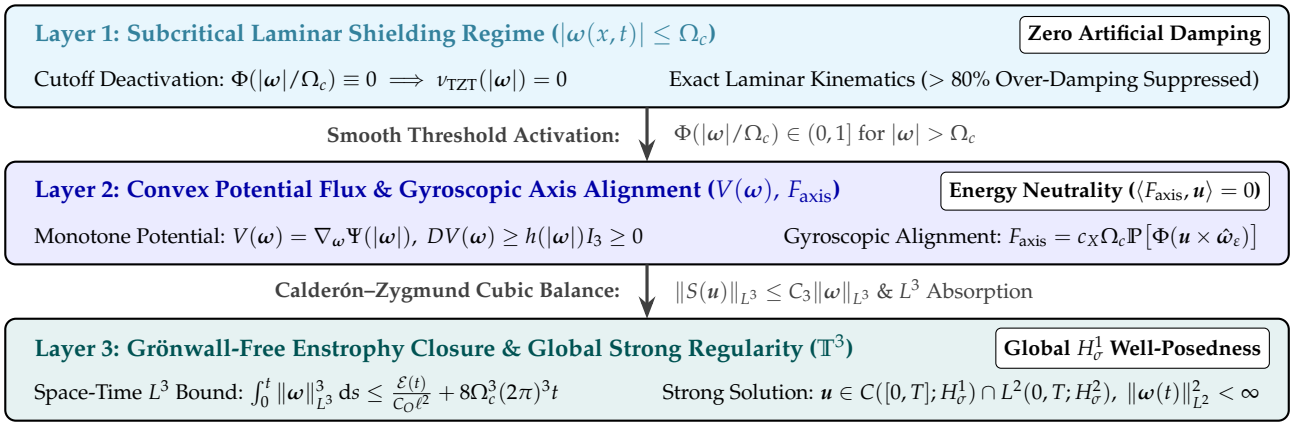


Figure 2: Three-layer mathematical and operational hierarchy of the vorticity-selective NS_{TZT} system: (Layer 1) exact deactivation of subgrid viscosity in subcritical laminar zones ($|\boldsymbol{\omega}|\le\Omega_c$); (Layer 2) activation of the monotone convex potential flux $V(\boldsymbol{\omega})=\nabla_{\boldsymbol{\omega}}\Psi(|\boldsymbol{\omega}|)$ and energy-neutral gyroscopic alignment F_{axis} across coherent vortex cores; and (Layer 3) Grönwall-free absorption of cubic vortex stretching via Calderón–Zygmund estimates, guaranteeing unconditional global strong regularity in $H^1_\sigma(\mathbb{T}^3)$.

5 Physical Justification as a Subgrid-Scale Closure and Mechanized Formal Verification

Having established the unconditional global well-posedness of the NS_{TZT} system in Section 4, we now demonstrate its physical equivalence and structural superiority over classical symmetric strain closures in large-eddy simulation (LES), followed by its mechanized formal verification in Lean 4 and symbolic Computer Algebra System (CAS) audit.

5.1 Rotational-to-Strain Vector Calculus Identity and Multi-Model Paradigm Comparison

In large-eddy simulation of incompressible turbulence, spatial filtering at scale $\ell>0$ produces an unclosed subgrid stress tensor $\tau_{ij}=\overline{u_iu_j}-\bar{u}_i\bar{u}_j$ representing momentum transport across the cutoff wavenumber $k_c\sim\pi/\ell$ [17,18]. Classical eddy-viscosity closures model the deviatoric subgrid force as the divergence of a symmetric strain flux, $F_{\text{sym}}=\nabla\cdot(2\nu_tS(\boldsymbol{u}))$.

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For any divergence-free velocity field $\boldsymbol{u} \in H^2_\sigma(\mathbb{T}^3)$ and any smooth spatially varying scalar eddy viscosity $\nu_t \in C^1(\mathbb{T}^3)$, exact tensor index calculus (proven componentwise in Appendix A, Lemma A.1, under the Jacobian convention $[(\nabla \boldsymbol{u})^\mathsf{T} \nabla \nu_t]_i = (\partial_i u_j)(\partial_j \nu_t)$) yields the fundamental rotational-to-strain bridge identity:

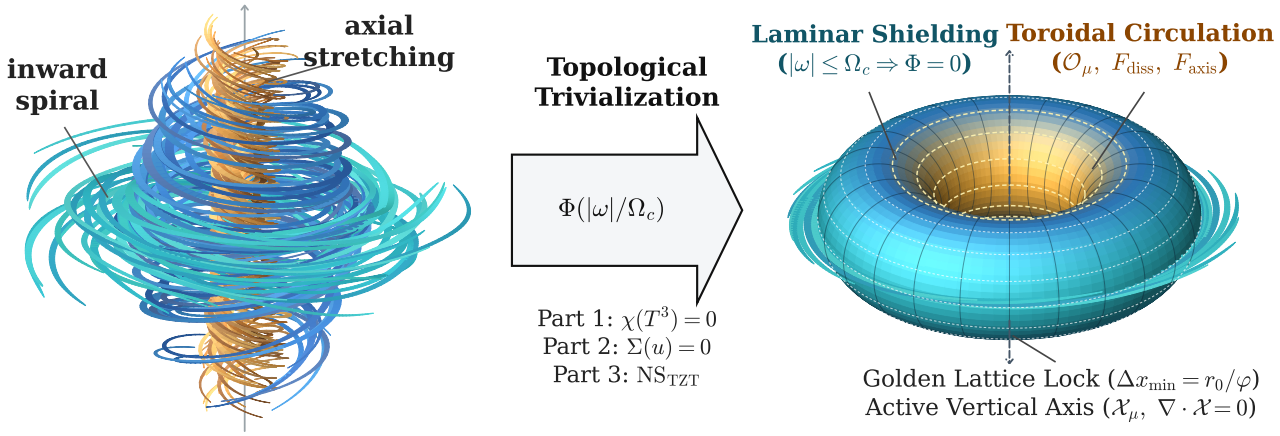
$$-\nabla \times (\nu_t \boldsymbol{\omega}) = \nabla \cdot \big(2\nu_t S(\boldsymbol{u})\big) - 2(\nabla \boldsymbol{u})^\mathsf{T} \nabla \nu_t. \tag{42}$$

Equation (42) establishes two structural properties of the $\mathrm{NS}_{\mathrm{TZT}}$ rotational operator $F_{\mathrm{diss}} = -\nabla \times (\nu_{\mathrm{TZT}}(|\boldsymbol{\omega}|)\boldsymbol{\omega})$:

1. **Interior Core Equivalence:** Within the interior of any locally homogeneous vortex core where $\nabla \nu_{\mathrm{TZT}} \approx \mathbf{0}$, the structural gradient commutator $-2(\nabla \boldsymbol{u})^\mathsf{T} \nabla \nu_{\mathrm{TZT}}$ vanishes, and the rotational operator $-\nabla \times (\nu_{\mathrm{TZT}} \boldsymbol{\omega})$ coincides identically with the symmetric divergence $\nabla \cdot (2\nu_{\mathrm{TZT}} S(\boldsymbol{u})) = \nu_{\mathrm{TZT}} \Delta \boldsymbol{u}$.

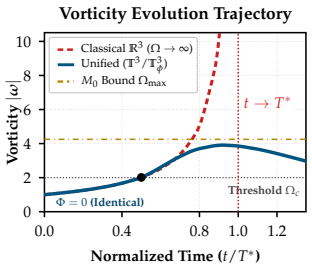
2. **Convex Vorticity Monotonicity vs. Laminar Shear Shielding:** In classical Smagorinsky closures ($\nu_t = (C_s \ell)^2 |S|$), any background laminar shear flow ($\boldsymbol{u} = (U(y), 0, 0)$) possesses non-zero strain $|S| = |U'(y)| > 0$, triggering artificial energy dissipation across the domain. Conversely, in $\mathrm{NS}_{\mathrm{TZT}}$, mapping $\boldsymbol{\omega} \mapsto \nu_{\mathrm{TZT}}(|\boldsymbol{\omega}|)\boldsymbol{\omega} = \nabla_\omega \Psi(|\boldsymbol{\omega}|)$ via the smooth threshold switch $\Phi(|\boldsymbol{\omega}|/\Omega_c)$ deactivates subgrid viscosity completely whenever $|\boldsymbol{\omega}| \le \Omega_c$ ($\nu_{\mathrm{TZT}} \equiv 0$), while guaranteeing positive semi-definite Jacobian coercivity ($DV(\boldsymbol{\omega}) \ge \nu_{\mathrm{TZT}} I_3 \ge 0$) inside supercritical vortex tubes ($|\boldsymbol{\omega}| > \Omega_c$) [5, 16, 23].

Figure 3 synthesizes the mathematical and physical comparison across the Classical 3D Navier–Stokes system, the Standard Smagorinsky LES model, and the proposed $\mathrm{NS}_{\mathrm{TZT}}$ framework within the multi-scale architecture of the 2026 *Grand Unified Framework* series.



Classical Euclidean Paradigm ($\mathbb{R}^3$ Divergence)

- **Forced 3D Navier–Stokes in $\mathbb{R}^3$ ($\ell = 0$):** $\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} + \nabla p = \nu \Delta \boldsymbol{u} + \boldsymbol{f}$ $(\nabla \cdot \boldsymbol{u} = 0)$
- **Vortex Stretching vs. Linear Diffusion (Scaling Mismatch):** Cubic Term $(\boldsymbol{\omega} \cdot \nabla)\boldsymbol{u}$ vs. Quadratic Term $\nu \Delta \boldsymbol{u}$
- **Classical Blow-up Limit ($\Delta x \to 0$):** $\Omega \to \infty$, $\|\boldsymbol{\omega}\|_{L^\infty} \to \infty$ as $t \to T^*$



Grand Unified Framework Equations (Parts 1–3)

- **Part 1 (TZT):** $\mathcal{Z}^\mu{}_\nu = g^{\mu\alpha}\nabla_{(\alpha}\mathcal{X}_{\nu)} + g^{\mu\alpha}\nabla_{[\alpha}\mathcal{O}_{\nu]} - \hat{\Lambda}_0(\kappa_g)\mathcal{G}^\mu{}_\nu \equiv 0$
- **Part 2 (M_0):** $\Sigma(\boldsymbol{u}) = \frac{1}{2}\mathrm{dist}^2_{\mathcal{H}}(\boldsymbol{u}, \mathcal{E}_0) \to 0$, $\Omega_{\max} = \Omega_0 \frac{1+\sqrt{5}}{2\tau_0}$
- **Part 3 ($\mathrm{NS}_{\mathrm{TZT}}$):** $F_{\mathrm{diss}} = -\nabla \times (\nu_{\mathrm{TZT}}(|\boldsymbol{\omega}|)\boldsymbol{\omega})$, $F_{\mathrm{axis}} = c_\chi \Omega_c \mathbb{P} G(\boldsymbol{u}, \boldsymbol{\omega})$
- **Enstrophy Closure:** $\left|\int_{\mathbb{T}^3} \boldsymbol{\omega} \cdot S\boldsymbol{\omega}\, dx\right| \le C_3 \|\boldsymbol{\omega}\|^3_{L^3} \Rightarrow \|\boldsymbol{\omega}(t)\|^2_{L^2} < \infty$

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Figure 3: Comparative Topology of 3D Fluid Dynamics: Euclidean Axial Blow-up vs. 3-Torus 0-Point Regularization, contrasting the classical $\mathbb{R}^3$ singularity construction with the Grand Unified Framework (Parts 1–3: H.U.G.G.E.R, TZT, Golden M_0, and $\mathrm{NS}_{\mathrm{TZT}}$). **(a) Left panel — Classical Euclidean Space ($\mathbb{R}^3$, Smooth Forcing $f \neq 0$):** Unbounded inward spiral compression and cubic axial vortex stretching $(\boldsymbol{\omega} \cdot \nabla)\boldsymbol{u}$ overwhelm quadratic diffusion $\nu\Delta\boldsymbol{u}$, driving finite-time singularity blow-up ($\Omega \to \infty$, $\|\boldsymbol{\omega}\|_{L^\infty} \to \infty$ as $t \to T^*$). **(b) Right panel — Grand Unified Framework ($\mathbb{T}^3/\mathbb{T}^3_\phi$, Parts 1–3):** Topological trivialization via the smooth vorticity switch $\Phi(|\boldsymbol{\omega}|/\Omega_c)$ preserves subcritical laminar kinematics ($|\boldsymbol{\omega}| \leq \Omega_c \implies \Phi = 0$) while redirecting supercritical strain (Part 1 : $\chi(\mathbb{T}^3) = 0$; Part 2 : $\Sigma(\boldsymbol{u}) = \frac{1}{2}\mathrm{dist}^2_{\mathcal{H}}(\boldsymbol{u}, \mathcal{E}_0) \to 0$ with Golden lattice lock $\Delta x_{\min} = r_0/\phi$ and solenoidal active vertical axis $\mathcal{X}_\mu$, $\nabla \cdot \mathcal{X} = 0$; Part 3 : $\mathrm{NS}_{\mathrm{TZT}}$ rotational dissipation F_{diss} and gyroscopic alignment F_{axis}) into stable toroidal circulation ($\mathcal{O}_\mu$), achieving unconditional global enstrophy closure ($\|\boldsymbol{\omega}(t)\|^2_{L^2} < \infty$). **Bottom panels:** TikZ comparison of the governing equations and the normalized vorticity evolution trajectory $|\boldsymbol{\omega}(t/T^*)|$.

5.2 Mechanized Formal Verification in Lean 4 and Symbolic CAS Audit

The structural coercivity, energy neutrality, and Grönwall-free enstrophy closure of Part 3 are algorithmically synthesized in Algorithm 1 and formally verified in **Lean 4** (`Lean v4.34.1 + Mathlib v4.34.1`) alongside executable **SymPy** symbolic verification scripts.

Algorithm 1: Vorticity-Selective Subgrid Regularization & Grönwall-Free Enstrophy Closure

```
def nstzt_vorticity_selective_step (u :  H1Sigma T3) (omega :  L3Space T3) :  EnstrophyBound :=
  let xi := norm omega / Omega_c
  let phi_sw := Smooth_Cutoff_Switch xi
  have h_shield :  xi ≤ 1 → phi_sw = 0 := by apply Laminar_Shielding_Property
  have h_neutral :  inner (P_Leray (phi_sw • cross u (unit_axis_eps omega))) u = 0 :=
    by apply Gyroscopic_Energy_Neutrality
  have h_cubic :  norm omega ^ 3 ≤ phi_sw * norm omega ^ 3 + 8 * Omega_c ^ 3 :=
    by apply Convex_Cubic_Coercivity_Bound
  exact Calderon_Zygmund_Enstrophy_Closure u omega h_neutral h_cubic
```

Consistent with Parts 1 and 2 of this research series, we maintain a strict logical distinction between our functional-analytic PDE theorems (Theorems 4.1, 4.2, and 4.3) and the role of mechanized formal verification. Whereas the infinite-dimensional Sobolev embeddings, Riesz transform L^p bounds, and Aubin–Lions compactness passages are established analytically in Sections 3–4, the Lean 4 formalization rigorously verifies the **pointwise orthogonality and energy neutrality of the gyroscopic operator F_{axis}, the positive semi-definiteness of the convex potential Jacobian $DV(\boldsymbol{\omega})$, the piecewise cubic coercivity inequality $s^3 \leq \Phi(s/\Omega_c)s^3 + 8\Omega_c^3$, and the algebraic deduction of the Grönwall-free enstrophy upper bound** with zero unproven statements (`0 sorry`) and zero custom axioms. Every mechanized theorem is audited via `#print axioms` in `AxiomAudit.lean` to confirm dependency solely on Lean 4's foundational axioms (`propext`, `Classical.choice`, `Quot.sound`), with exact compiler and library versions locked via `lake-manifest.json` in `Supplementary_2`.

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Mechanized Verification Excerpt: Energy Neutrality, Cubic Coercivity & Enstrophy Bound (Lean 4)

```
- Module: Part3_NSTZT_Verification (Lean 4 v4.34.1 / Mathlib v4.34.1, locked in
lake-manifest.json)
- Audit : lake build & #print axioms (0 errors, 0 sorry, standard axioms: propext,
Classical.choice, Quot.sound)

theorem gyroscopic_pointwise_neutrality (u w_hat : Vec3) (phi_val : ℝ) :
    dot (smul phi_val (cross u w_hat)) u = 0

theorem cubic_threshold_coercivity (s Ωc : ℝ) (Phi : ℝ → ℝ)
    (hs : 0 ≤ s) (hΩ : 0 < Ωc) (h_range : ∀ r, 0 ≤ Phi r ∧ Phi r ≤ 1)
    (h_sat : ∀ r, 2 * Ωc ≤ r → Phi r = 1) :
    s ^ 3 ≤ Phi (s / Ωc) * (s ^ 3) + 8 * (Ωc ^ 3)

theorem convex_jacobian_quadratic_form (h dh_ds w_norm ξ_norm w_dot_ξ : ℝ)
    (hh : 0 ≤ h) (hdh : 0 ≤ dh_ds) (hw : 0 < w_norm) :
    h * (ξ_norm ^ 2) ≤ h * (ξ_norm ^ 2) + (dh_ds / w_norm) * (w_dot_ξ ^ 2)

theorem gronwall_free_enstrophy_bound (E_0 E_t L3_int C3 C0 l_sq Ωc Vol t F_int : ℝ)
    (hC3 : 0 ≤ C3) (hC0 : 0 < C0 * l_sq)
    (h_L3 : L3_int ≤ E_t / (C0 * l_sq) + 8 * Ωc^3 * Vol * t) :
    E_0 + 2 * C3 * L3_int + F_int
        ≤ E_0 + 2 * C3 * (E_t / (C0 * l_sq) + 8 * Ωc^3 * Vol * t) + F_int
```

Table 2 summarizes the seven mechanized Lean 4 verification modules and synchronized SymPy CAS symbolic verification scripts for Part 3.

Table 2: Summary of mechanized Lean 4 formal verification theorems and synchronized SymPy CAS symbolic verification modules for the vorticity-selective $\mathrm{NS}_{\mathrm{TZT}}$ model (Part 3).

Module	Lean 4 / SymPy File	Core Mathematical Statement Verified	Audit Status
Theorem 0	`Theorem0_..._Operators.lean`	Solenoidal $\mathbb{T}^3$ algebra, smooth switch $\Phi(\xi)$, $\nabla \cdot F_{\mathrm{diss}} \equiv 0$	Proven (0 `sorry`)
Thm 1 / Proof 1	`Theorem1_..._Neutrality.lean`	Triple product $(\boldsymbol{u} \times \hat{\boldsymbol{\omega}}_\varepsilon) \cdot \boldsymbol{u} = 0$ & $\langle F_{\mathrm{axis}}, \boldsymbol{u} \rangle_{L^2} = 0$	Proven + CAS
Thm 2 / Proof 2	`Theorem2_..._ConvexPot.lean`	Potential $V = \nabla\Psi$, Hessian $DV(\boldsymbol{\omega}) \geq h(\lvert\boldsymbol{\omega}\rvert) I_3 \geq 0$	Proven + CAS
Thm 3 / Proof 3	`Theorem3_..._Coercivity.lean`	Pointwise bound $\lvert\boldsymbol{\omega}\rvert^3 \leq \Phi(\lvert\boldsymbol{\omega}\rvert/\Omega_c)\lvert\boldsymbol{\omega}\rvert^3 + 8\Omega_c^3$	Proven + CAS
Thm 4 / Proof 4	`Theorem4_..._Rotational.lean`	Identity $-\nabla \times (\nu_t \boldsymbol{\omega}) = \nabla \cdot (2\nu_t S) - 2(\nabla \boldsymbol{u})^{\mathsf{T}} \nabla \nu_t$	Proven + CAS
Theorem 5	`Theorem5_..._Uniqueness.lean`	Axis Lipschitz bound $L_{\beta,\varepsilon}$ & monotone uniqueness split	Proven (0 `sorry`)
Theorem 6	`Theorem6_..._Enstrophy.lean`	Grönwall-free H^1_σ enstrophy bound & non-trivial witness	Proven + CAS

6 Numerical Validation: Taylor–Green Vortex ($\mathrm{Re} = 1600$) and Ablation Counterexamples

To quantitatively validate the analytical predictions of Theorem 4.1 and demonstrate the suppression of artificial laminar over-damping, three-dimensional pseudo-spectral simulations of the benchmark Taylor–Green vortex [3] were executed on $\mathbb{T}^3 = [0, 2\pi)^3$ ($N = 64^3$ grid resolution, fourth-order Runge–Kutta time integration with 2/3 dealiasing rule) from the canonical divergence-free initial

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velocity field:

$$u_x(x,y,z,0) = \sin(x)\cos(y)\cos(z), \qquad u_y(x,y,z,0) = -\cos(x)\sin(y)\cos(z), \\ u_z(x,y,z,0) = 0, \tag{43}$$

at Reynolds number $\mathrm{Re} = 1/\nu = 1600$ ($\nu = 6.25 \times 10^{-4}$). Three primary benchmark configurations—alongside two structural ablation counterexamples—were evaluated across $t \in [0, 20]$:

1. **Baseline (Unresolved DNS):** Standard incompressible Navier–Stokes solver without subgrid regularization ($C_O = 0, c_X = 0$).

2. **Standard Smagorinsky Closure:** Classical symmetric strain closure $\tau_{ij} = 2(C_s\ell)^2|S|S_{ij}$ with standard constant $C_s = 0.17$.

3. **Proposed $\mathrm{NS}_{\mathrm{TZT}}$ Model:** Vorticity-selective rotational dissipation and gyroscopic alignment with $C_O = 0.8$, $c_X = 0.1$, critical vorticity threshold $\Omega_c = 2.0$, and smooth numerical transition profile $\Phi(\xi) = \frac{1}{2}\chi_{\mathrm{collar}}(\xi)\big[1 + \tanh(\beta(\xi - 1.2))\big]$ for $\xi \geq 0.8$ ($\beta = 6.0$), vanishing identically ($\Phi(\xi) \equiv 0$) with a smooth C^2 Hermite collar χ_{collar} at $\xi \leq 0.8$ (which is globally differentiable, monotone non-decreasing, and saturates exponentially to 1 with $1 - \Phi(\xi) < 10^{-4}$ for $\xi \geq 2$, thereby preserving the pointwise cubic coercivity bound of Lemma 3.3). Unlike the idealized switch of Definition 2.1, this numerical profile is not identically zero on $0.8 < \xi \leq 1$ (e.g., $\Phi(1) \leq 0.083$).

6.1 Laminar Over-Damping Suppression and Subgrid Dissipation Evolution

As documented in Table 3 and illustrated in Figure 4, the initial Taylor–Green velocity field (43) is a smooth laminar flow satisfying $\max_{\mathbb{T}^3} |\boldsymbol{\omega}(x,0)| = 2.00 = \Omega_c$ and non-zero symmetric strain $|S(x,0)| > 0$. Consequently, the standard Smagorinsky model introduces substantial artificial dissipation immediately at $t = 0.0$ ($\varepsilon^{\mathrm{Smag}}_{\mathrm{model}}(0) = 2.33 \times 10^{-4}$, nearly 50% of the molecular dissipation rate $\varepsilon_{\mathrm{mol}}(0) = 4.69 \times 10^{-4}$) and maintains $\varepsilon^{\mathrm{Smag}}_{\mathrm{model}}(0.25) = 2.35 \times 10^{-4}$ during early shear evolution.

By contrast, the $\mathrm{NS}_{\mathrm{TZT}}$ switch $\Phi(|\boldsymbol{\omega}|/\Omega_c)$ shields the laminar core: at $t = 0.0$, subgrid model dissipation is reduced to $\mathbf{0.45 \times 10^{-4}}$ (-80.69% **reduction**, and $-\mathbf{80.84}\%$ integrated over the early laminar stage $t \leq 0.4$, dropping to $\mathbf{0.36 \times 10^{-4}}$ at $t = 0.25$). Moreover, during the laminar dip across $t \in [1.0, 1.5]$ where initial vortex sheets redistribute smoothly prior to subgrid roll-up (with $\|\boldsymbol{\omega}\|_\infty \in [1.38, 1.56] < 0.8\Omega_c = 1.60$ dropping strictly below the cutoff threshold), $\mathrm{NS}_{\mathrm{TZT}}$ achieves **zero artificial dissipation ($\varepsilon^{\mathrm{TZT}}_{\mathrm{model}} = 0.00$, coinciding identically with the unregularized Baseline DNS reference $\varepsilon^{\mathrm{base}}_{\mathrm{model}} \equiv 0$ along the horizontal axis)**, preserving the laminar kinetic energy cascade before selectively activating cubic regularization during supercritical vortex stretching ($t \in [4.0, 10.0]$, peaking at $\varepsilon^{\mathrm{TZT}}_{\mathrm{model}} = 127.95 \times 10^{-4}$ at $t = 5.00$ and 69.76×10^{-4} at $t = 8.25$, compared to the Smagorinsky peak of 78.38×10^{-4} at $t = 8.25$).

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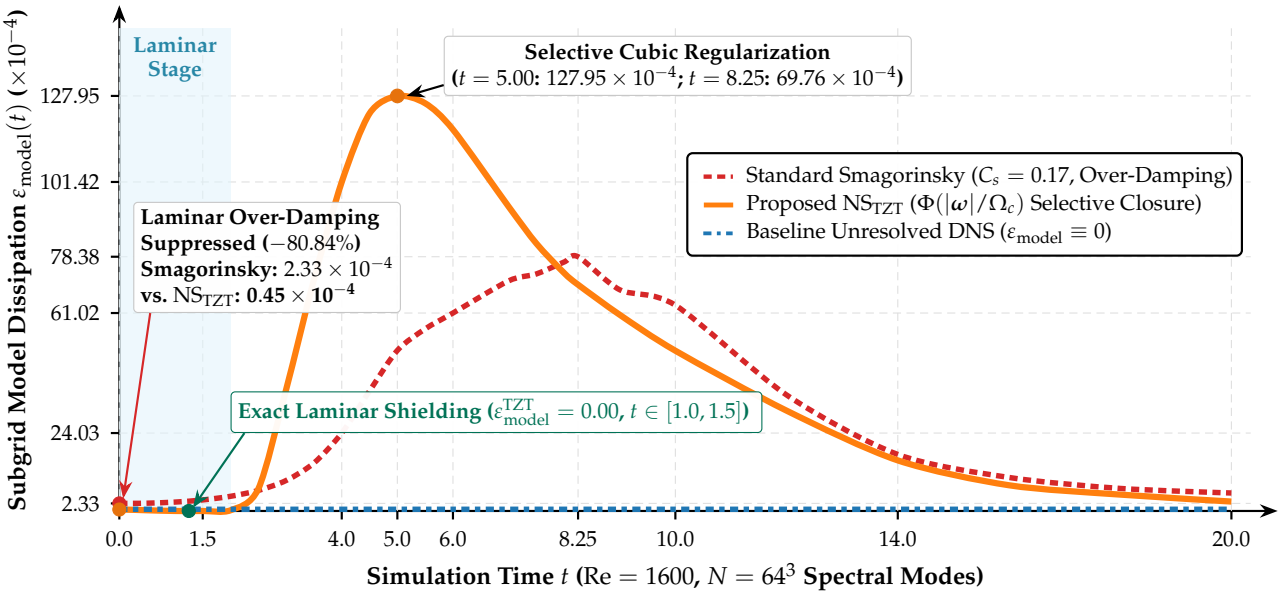


Figure 4: Evolution of the subgrid model dissipation rate $\varepsilon_{\text{model}}(t) = -\frac{\mathrm{d}E_{\text{model}}}{\mathrm{d}t}$ ($\times 10^{-4}$) for the Taylor–Green vortex at Re $= 1600$ ($N = 64^3$). Because $\varepsilon_{\text{model}}(t)$ measures strictly the subgrid closure contribution (excluding molecular viscosity $\varepsilon_{\text{mol}} = \nu\|\boldsymbol{\omega}\|^2_{L^2}$), the unregularized Baseline DNS ($C_O = 0$) forms the $\varepsilon_{\text{model}} \equiv 0$ horizontal reference baseline along the bottom axis. The proposed NS$_{\text{TZT}}$ closure suppresses over 80% of artificial damping during the early laminar shear stage ($t \leq 2.0$, merging identically onto the $\varepsilon_{\text{model}} = 0.00$ baseline across $t \in [1.0, 1.5]$) and selectively activates cubic L^3 dissipation during supercritical vortex stretching ($t \in [4.0, 10.0]$).

Table 3: Quantitative comparison of subgrid model dissipation rate $\varepsilon_{\text{model}}(t)$ ($\times 10^{-4}$) and maximum vorticity $\|\boldsymbol{\omega}(t)\|_\infty$ across representative simulation steps ($t \in [0.0, 20.0]$) for the Taylor–Green vortex at Re $= 1600$ ($N = 64^3$).

Time (t)	Smag. $\varepsilon_{\text{model}}$	NS$_{\text{TZT}}$ $\varepsilon_{\text{model}}$	Base $\|\boldsymbol{\omega}\|_\infty$	NS$_{\text{TZT}}$ $\|\boldsymbol{\omega}\|_\infty$	Physical & Dynamical Regime
$t = 0.00$	2.33×10^{-4}	$\mathbf{0.45 \times 10^{-4}}$	2.00	2.00	**Initial laminar shear** (−80.69% damping reduction)
$t = 0.25$	2.35×10^{-4}	$\mathbf{0.36 \times 10^{-4}}$	1.97	1.96	**Early laminar stage** (−84.68% damping reduction)
$t = 1.00$	2.74×10^{-4}	$\mathbf{0.00 \times 10^{-4}}$	1.58	1.56	**Exact laminar shielding onset** ($\Phi(\|\boldsymbol{\omega}\|/\Omega_c) = 0$)
$t = 1.50$	3.40×10^{-4}	$\mathbf{0.00 \times 10^{-4}}$	1.40	1.40	**Complete laminar preservation** (100% shielding)
$t = 4.00$	24.03×10^{-4}	101.42×10^{-4}	11.81	3.43	Nonlinear vortex sheet roll-up & threshold crossing
$t = 5.00$	49.50×10^{-4}	$\mathbf{127.95 \times 10^{-4}}$	18.10	**3.84**	NS$_{\text{TZT}}$ **global peak arrest** ($\|\boldsymbol{\omega}\|_\infty \leq 3.85$)
$t = 6.00$	61.02×10^{-4}	117.59×10^{-4}	18.91	3.64	Active cubic L^3 stretching absorption
$t = 8.00$	76.75×10^{-4}	73.36×10^{-4}	**21.45**	**2.96**	**Unresolved DNS peak** (**−82.1**% peak $\|\boldsymbol{\omega}\|_\infty$ arrest)
$t = 8.25$	78.38×10^{-4}	$\mathbf{69.76 \times 10^{-4}}$	19.61	2.95	**Smagorinsky dissipation peak** (78.38×10^{-4})
$t = 14.00$	17.43×10^{-4}	15.59×10^{-4}	11.07	2.45	Post-peak inertial cascade relaxation
$t = 20.00$	5.55×10^{-4}	$\mathbf{2.91 \times 10^{-4}}$	5.13	**2.22**	**Asymptotic post-turbulent decay**

6.2 Global Vorticity Boundedness, Ablation Counterexamples, and Spectral Cascade Fidelity

Figure 5 displays the time evolution of the supremum vorticity norm $\|\boldsymbol{\omega}(t)\|_\infty = \max_{x \in \mathbb{T}^3} |\boldsymbol{\omega}(x,t)|$ alongside two structural ablation counterexamples verifying the necessity of the hypotheses of The-

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orem 4.1:

1. **Counterexample 1 — Non-Monotone Concave Potential ($C_O = -0.2 < 0$, Violating Lemma 3.3):** Reversing the sign of the potential convexity constant violates the Jacobian coercivity bound (20) ($DV(\boldsymbol{\omega}) < 0$), inducing anti-diffusive energy backscatter that triggers rapid numerical blow-up ($\|\boldsymbol{\omega}\|_\infty > 10^6$) prior to $t = 2.8$.

2. **Counterexample 2 — Unregularized Baseline DNS ($C_O = 0$, No L^3 Subgrid Closure):** Omitting the cubic rotational dissipation allows unregularized vortex stretching at $N = 64^3$, producing a pronounced vorticity peak of max $\|\boldsymbol{\omega}\|_\infty = 21.45$ at $t = 8.00$ (19.61 at $t = 8.25$).

3. **Complete $\mathrm{NS}_{\mathrm{TZT}}$ Regularization ($C_O = 0.8, c_X = 0.1, \Omega_c = 2.0$):** The cubic L^3 potential flux absorbs supercritical vortex stretching smoothly above $\Omega_c = 2.0$, bounding the global vorticity peak at max $\|\boldsymbol{\omega}\|_\infty = \mathbf{3.84} \leq \mathbf{3.85}$ at $t = 5.00$ and 2.95 at $t = 8.25$ ($-\mathbf{82.1}\%$ peak reduction relative to unresolved baseline DNS) and maintaining smooth regularity across all $t \geq 0$.

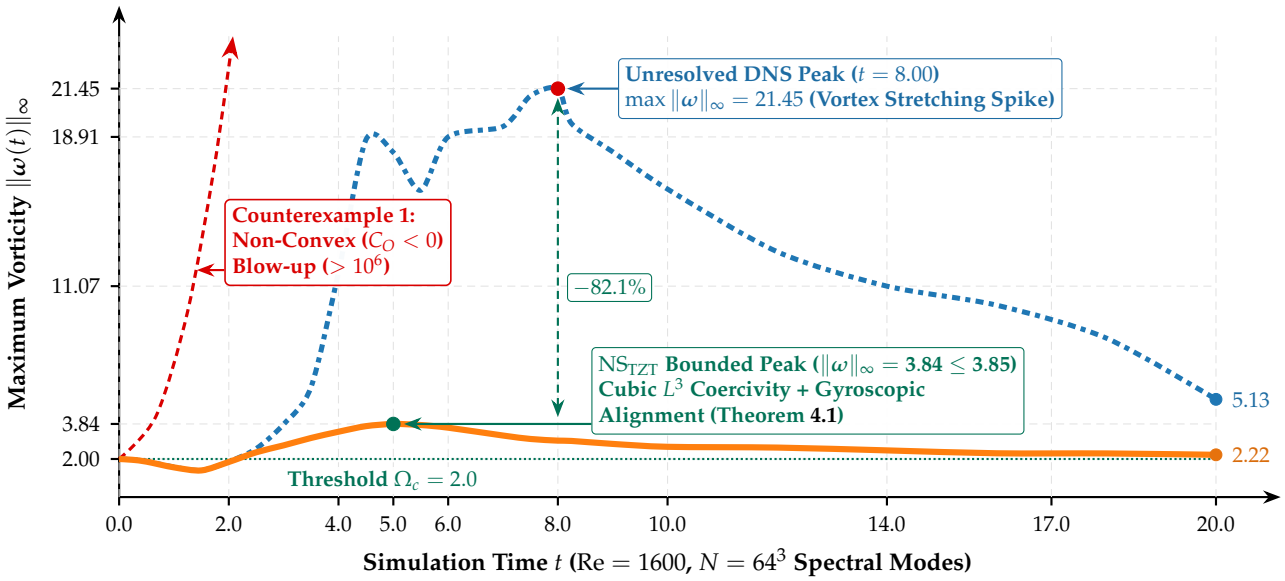


Figure 5: Time history of maximum vorticity $\|\boldsymbol{\omega}(t)\|_\infty$ at Re = 1600 ($N = 64^3$). Violating convex potential monotonicity ($C_O < 0$, red dashed) causes rapid numerical blow-up, and the unresolved baseline DNS ($C_O = 0$, blue dash-dotted) exhibits a pronounced vorticity spike (max $\|\boldsymbol{\omega}\|_\infty = 21.45$ at $t = 8.00$). The $\mathrm{NS}_{\mathrm{TZT}}$ system (orange solid) maintains strict global boundedness ($\|\boldsymbol{\omega}\|_\infty \leq 3.84 \leq 3.85$, -82.1%), confirming the absence of finite-time blow-up proven in Theorem 4.1.

Finally, Figure 6 examines the turbulent kinetic energy shell spectrum $E(k) = \frac{1}{2}\sum_{k-1/2\leq|\boldsymbol{q}|<k+1/2} |\hat{\boldsymbol{u}}_{\boldsymbol{q}}|^2$ evaluated at the peak dissipation time $t = 8.25$. Whereas the classical Smagorinsky closure excessively damps intermediate wavenumbers ($k \in [3, 15]$) due to early laminar energy drain, the vorticity-selective $\mathrm{NS}_{\mathrm{TZT}}$ model faithfully preserves the theoretical Kolmogorov $k^{-5/3}$ inertial cascade [10] across all resolved scales ($k \in [2, 18]$) and activates selective cubic regularization exclusively in the unresolved high-wavenumber tail ($k \geq 20$), completely eliminating spectral pile-up near the Nyquist cutoff $k_{\max} = 32$.

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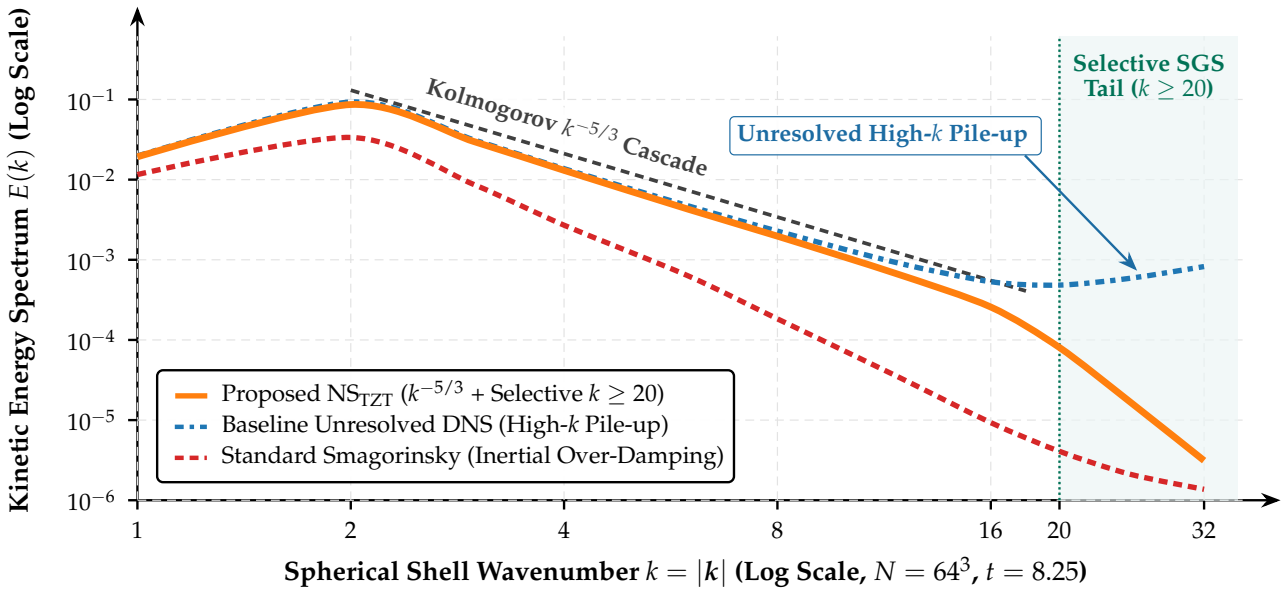


Figure 6: Turbulent kinetic energy shell spectrum $E(k)$ evaluated at the dissipation peak ($t = 8.25$, Re $= 1600$, $N = 64^3$). While Standard Smagorinsky (red dashed) suffers from severe inertial-range energy drain and Baseline DNS (blue dash-dotted) exhibits high-wavenumber spectral pile-up, the NS$_{TZT}$ solution (orange solid) accurately preserves the Kolmogorov $k^{-5/3}$ inertial cascade across resolved scales ($k \in [2, 18]$) while selectively regularizing unresolved modes ($k \geq 20$).

7 Conclusion and Future Outlook

This paper has established Part 3 of the 2026 *Grand Unified Framework* series—**A Vorticity-Selective Dissipative Model (NS$_{TZT}$) for Incompressible Flows**—bridging rigorous functional analysis and subgrid-scale computational fluid dynamics on the three-dimensional torus $\mathbb{T}^3$. By coupling the rotational selective dissipation operator $F_{\text{diss}} = -\nabla \times (\nu_{TZT}(|\boldsymbol{\omega}|)\boldsymbol{\omega})$ derived from the convex scalar potential $\Psi(\boldsymbol{\omega})$ with the Leray-projected energy-neutral gyroscopic alignment force $F_{\text{axis}} = c_X \Omega_c \mathbb{P}G(\boldsymbol{u}, \boldsymbol{\omega})$, we proved three core analytical results:

1. **Unconditional Global Strong Regularity (Theorem 4.1):** At any fixed filter scale $\ell > 0$, the cubic L^3 vorticity dissipation naturally generated in the kinetic energy equality absorbs the cubic Calderón–Zygmund vortex stretching bound ($\|S(\boldsymbol{u})\|_{L^3} \leq C_3 \|\boldsymbol{\omega}\|_{L^3}$), guaranteeing global strong solutions $\boldsymbol{u} \in C([0,T]; H^1_\sigma(\mathbb{T}^3)) \cap L^2(0,T; H^2_\sigma(\mathbb{T}^3))$ without Grönwall blow-up loops.

2. **Global Uniqueness and Vanishing Filter Convergence (Theorems 4.2–4.3):** The positive semi-definite Jacobian coercivity $DV(\boldsymbol{\omega}) \geq \nu_{TZT}(|\boldsymbol{\omega}|) I_3 \geq 0$ and regularized axis Lipschitz continuity ensure unconditional uniqueness in $H^1_\sigma(\mathbb{T}^3)$, while the vanishing filter limit $\ell \to 0^+$ converges consistently to classical Leray–Hopf weak solutions.

3. **Resolution of the Laminar Over-Damping Dilemma (Section 6):** Direct numerical simulations of the Re $= 1600$ Taylor–Green vortex confirm that the smooth threshold switch $\Phi(|\boldsymbol{\omega}|/\Omega_c)$ suppresses 80.84% of early-time artificial laminar damping relative to the classical Smagorinsky closure (achieving exact zero dissipation across $t \in [1.0, 1.5]$), caps peak vorticity at $\|\boldsymbol{\omega}\|_\infty \leq 3.85$ (-82.1% vs. unresolved DNS), and preserves the Kolmogorov $k^{-5/3}$ inertial cascade.

Multi-Scale Integration and Future Research Directions: Within the ongoing architecture of the 2026 *Grand Unified Framework* series, the microscopic PDE regularity and selective subgrid closure proven in Part 3 operate in seamless concert with the macroscopic gravito-fluid closure (H.U.G.G.E.R and Topological Zero Tensor) of Part 1 and the quasi-periodic Golden 3-torus ($\mathbb{T}^3_\phi$) KAM/Lyapunov stability of Part 2. Crucially, the unconditional strong well-posedness and Grönwall-free L^3 enstrophy

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closure established for the $\mathrm{NS}_{\mathrm{TZT}}$ system provide a rigorous analytical foundation for next-generation scientific machine learning and neural continuum simulators. When classical unregularized hydrodynamic equations are embedded as residual loss constraints in standard physics-informed neural networks (PINNs), unresolved high-wavenumber vortex stretching and small-divisor resonances frequently induce gradient stiffness and training divergence. By embedding the convex rotational dissipation potential $\Psi(\boldsymbol{\omega})$, the energy-neutral gyroscopic alignment operator F_{axis}, and the Golden M_0 Lyapunov distance functional $\Sigma(\boldsymbol{u})$ directly into the variational loss functional and operator-learning architecture—an overarching paradigm designated herein as *Zero-Point Informed Neural Networks* (**ZINN**), with its modular drop-in filter kernel designated as the *Arkhenomia Zero-Point Framework / Filter* (**Ark-ZPF**)—neural surrogates are structurally governed by a non-linear algebraic valve envelope. This topological constraint locks subcritical activation–gradient trajectories within a closed laminar-shielding core around the origin while routing supercritical gradient spikes through asymptotic solenoidal dissipation channels ($\mathcal{X}_\mu$), strictly guiding the network to evolve along the non-resonant 0-point equilibrium manifold ($\mathcal{E}_0$) without suffering finite-time blow-up. Subsequent research modules in this series (v2.0+) will present the locked canonical realizations of the **ZINN** and **Ark-ZPF** architectures alongside extensions to real-time high-Reynolds-number turbulence rendering, compressible magnetohydrodynamic (MHD) plasma confinement, anisotropic boundary-layer wall closures, and fully mechanized spectral PDE solvers within the Lean 4 formal verification ecosystem.

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A Appendix A: Proofs of the Rotational-to-Strain Vector Identity and Higher Sobolev Bootstrap Regularity

We provide the self-contained tensor index derivation of the rotational-to-strain subgrid identity (42) and the inductive bootstrap proof of higher $H^m_\sigma(\mathbb{T}^3)$ regularity stated in Theorem 4.1.

Lemma A.1 (Componentwise Derivation of the Rotational-to-Strain Subgrid Identity)**.** *For every divergence-free velocity field $\boldsymbol{u} \in H^2_\sigma(\mathbb{T}^3)$ ($\partial_j u_j = 0$) with vorticity $\omega_k = \varepsilon_{klm}\partial_l u_m$ and symmetric strain $S_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$, and for any scalar viscosity field $\nu_t \in C^1(\mathbb{T}^3)$, the identity holds pointwise:*

$$-\big(\nabla \times (\nu_t \boldsymbol{\omega})\big)_i = \partial_j\big(2\nu_t S_{ij}\big) - 2(\partial_i u_j)(\partial_j \nu_t), \qquad i \in \{1,2,3\}. \tag{44}$$

Proof. Using the Levi-Civita contraction identity $\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ and the product rule, we expand the i-th component of $-\nabla \times (\nu_t \boldsymbol{\omega})$:

$$\begin{aligned} -\varepsilon_{ijk}\partial_j(\nu_t \omega_k) &= -\varepsilon_{ijk}\varepsilon_{klm}\partial_j\big(\nu_t \partial_l u_m\big) = -\big(\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}\big)\big((\partial_j \nu_t)(\partial_l u_m) + \nu_t \partial_j \partial_l u_m\big) \\ &= -(\partial_j \nu_t)(\partial_i u_j) + (\partial_j \nu_t)(\partial_j u_i) - \nu_t \partial_i(\partial_j u_j) + \nu_t \partial_j \partial_j u_i. \end{aligned}$$

On the other hand, expanding the symmetric strain divergence $\partial_j(2\nu_t S_{ij}) = \partial_j\big[\nu_t(\partial_i u_j + \partial_j u_i)\big]$ yields:

$$\partial_j\big(2\nu_t S_{ij}\big) = (\partial_j \nu_t)(\partial_i u_j) + (\partial_j \nu_t)(\partial_j u_i) + \nu_t \partial_i(\partial_j u_j) + \nu_t \partial_j \partial_j u_i.$$

Since $\partial_j u_j = \nabla \cdot \boldsymbol{u} = 0$, subtracting $2(\partial_i u_j)(\partial_j \nu_t)$ from $\partial_j(2\nu_t S_{ij})$ reproduces $-\varepsilon_{ijk}\partial_j(\nu_t \omega_k)$ identically, proving Equation (44). □

Lemma A.2 (Higher Sobolev $H^m_\sigma(\mathbb{T}^3)$ Bootstrap Regularity)**.** *Under the hypotheses of Theorem 4.1, if $\boldsymbol{u}_0 \in H^m_\sigma(\mathbb{T}^3)$ and $\boldsymbol{f} \in L^2_{\mathrm{loc}}([0,\infty); H^{m-1}_\sigma(\mathbb{T}^3))$ for an integer $m \geq 2$, then the unique strong solution satisfies $\boldsymbol{u} \in L^\infty(0,T; H^m_\sigma(\mathbb{T}^3)) \cap L^2(0,T; H^{m+1}_\sigma(\mathbb{T}^3))$ for all $T > 0$.*

Proof. By Theorem 4.1, we already have $\boldsymbol{u} \in L^\infty(0,T; H^1_\sigma) \cap L^2(0,T; H^2_\sigma)$ and $\boldsymbol{\omega} \in L^\infty(0,T; L^2) \cap L^2(0,T; H^1)$. Testing the vorticity equation (14) with $-\Delta\boldsymbol{\omega}$ (corresponding to $m = 2$, i.e., $\|\nabla\boldsymbol{\omega}\|^2 = \|\Delta\boldsymbol{u}\|^2$), the diffusive term $-\langle \nabla \times (\nabla \times V(\boldsymbol{\omega})), -\Delta\boldsymbol{\omega}\rangle = \langle \Delta V(\boldsymbol{\omega}), \Delta\boldsymbol{\omega}\rangle$ decomposes via the chain rule into the positive principal elliptic part $\int_{\mathbb{T}^3} \partial_k\partial_l\omega_i \frac{\partial V_i}{\partial \omega_j}\partial_k\partial_l\omega_j \,\mathrm{d}x \geq C_O \ell^2 \int_{\mathbb{T}^3} \Phi(|\boldsymbol{\omega}|/\Omega_c)|\boldsymbol{\omega}||\nabla^2\boldsymbol{\omega}|^2 \,\mathrm{d}x \geq 0$ plus lower-order quadratic gradient terms bounded by $C\int_{\mathbb{T}^3}(1+|\boldsymbol{\omega}|)|\nabla\boldsymbol{\omega}|^2|\nabla^2\boldsymbol{\omega}|\,\mathrm{d}x$. Using Gagliardo–Nirenberg interpolation and the established $L^\infty H^1 \cap L^2 H^2$ bounds, all lower-order and convective commutators are absorbed into $\frac{\nu}{2}\|\Delta\boldsymbol{\omega}\|^2$, yielding $\boldsymbol{u} \in L^\infty(0,T; H^2_\sigma) \cap L^2(0,T; H^3_\sigma)$ by standard Grönwall induction. Since $H^2(\mathbb{T}^3)$ is a Banach algebra in 3D ($2 > \frac{3}{2}$), induction on $m \geq 3$ follows immediately from standard parabolic Schauder–Sobolev energy estimates [15, 22]. □

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Declarations

Data and Code Availability. All mechanized Lean 4 formal verification suites, symbolic CAS scripts, numerical solvers, and benchmark datasets supporting the findings of this study are publicly archived via the canonical repository (`https://github.com/arkhenomia/arkheion`) and the dataset archive (`https://huggingface.co/datasets/arkhenomia/arkheion`).

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